

TRM V1 — Section 1

Document Overview & Traceability Scope

Document: Traceability Requirements Matrix (TRM V1)

Project:

TODP (Transparent Opportunity Decision Platform)

Document Type:

Traceability Requirements Matrix

Version:

V1

Status:

APPROVED

Classification:

Architecture Governance Artifact

Parent Documents:

- PRD V1 (Frozen)

- ADD V1 (Frozen)

- DDD V1 (Frozen)

- SDD V1 (Frozen)

- IAD V1 (Frozen)

- TDD V1 (Frozen)

1.1 Purpose

The Traceability Requirements Matrix (TRM) establishes end-to-end traceability across all approved TODP architecture and design documents.

The purpose of the TRM is to:

- Ensure requirement coverage
- Ensure implementation alignment
- Ensure architectural consistency
- Ensure testing completeness
- Ensure governance compliance
- Ensure auditability and traceability

The TRM serves as the authoritative linkage document connecting business requirements to technical implementation.

1.2 Traceability Objectives

TRM-OBJ-01

Provide complete requirement visibility.

TRM-OBJ-02

Verify architecture coverage.

TRM-OBJ-03

Verify design coverage.

TRM-OBJ-04

Verify implementation readiness.

TRM-OBJ-05

Verify test coverage.

TRM-OBJ-06

Support governance reviews.

1.3 Scope

This TRM covers traceability between:

PRD

↓

ADD

↓

DDD

↓

SDD

↓

IAD

↓

TDD

↓

Implementation

↓

Testing

All approved requirements within TODP V1 fall within the scope of this document.

1.4 Traceability Layers

Layer 1

Business Requirements

(PRD)

Layer 2

Architecture Requirements

(ADD)

Layer 3

Data Requirements

(DDD)

Layer 4

Solution Requirements

(SDD)

Layer 5

Interface Requirements

(IAD)

Layer 6

Technical Requirements

(TDD)

Layer 7

Implementation Requirements

(Code)

Layer 8

Validation Requirements

(Testing)

1.5 Traceability Principle

Every approved requirement must have:

Source

↓

Design Mapping

↓

Implementation Mapping

↓

Verification Mapping

No requirement may exist without traceability.

1.6 Traceability Model

Requirement



Architecture



Design



Interface



Implementation



Testing



Verification

1.7 Traceability Benefits

Benefits include:

Requirement Visibility

Coverage Validation

Impact Analysis

Change Management

Governance Compliance

Audit Readiness

1.8 Governance Importance

The TRM acts as the primary governance artifact used to verify that:

Business requirements are implemented.

Architectural requirements are respected.

Technical requirements are enforced.

Testing requirements are fulfilled.

1.9 Stakeholders

Primary Stakeholders:

Product Owner

Architect

Technical Lead

Developers

QA Engineers

Operations Team

Governance Reviewers

Auditors

1.10 Document Dependencies

This document depends upon:

PRD V1

ADD V1

DDD V1

SDD V1

IAD V1

TDD V1

All referenced documents must remain version controlled.

1.11 Traceability Coverage Domains

Coverage includes:

Authentication

Authorization

Opportunity Management

Authority Management

Evidence Management

Retrieval

Ranking

Recommendation

Governance

Auditability

Security

Operations

1.12 Requirement Categories

Functional Requirements

Architectural Requirements

Data Requirements

Interface Requirements

Technical Requirements

Operational Requirements

Security Requirements

Governance Requirements

1.13 Traceability Matrix Structure

Each matrix entry contains:

Requirement ID

Requirement Description

Source Document

Target Document

Implementation Reference

Verification Reference

Status

1.14 Requirement Lifecycle

Requirement Created



Requirement Approved



Requirement Designed



Requirement Implemented



Requirement Tested



Requirement Verified



Requirement Maintained

1.15 Requirement Status Values

PROPOSED

APPROVED

IMPLEMENTED

TESTED

VERIFIED

RETIRED

1.16 Verification Principles

Every requirement must be:

Traceable

Testable

Verifiable

Auditable

Reproducible

1.17 Change Management Support

The TRM supports:

Impact Analysis

Requirement Changes

Architecture Changes

Implementation Changes

Testing Changes

Governance Changes

1.18 Auditability Requirements

Every traceability record must support:

Requirement Audit

Design Audit

Implementation Audit

Testing Audit

Governance Audit

1.19 Traceability Requirements

Every requirement must support:

Forward Traceability

Backward Traceability

Bidirectional Traceability

1.20 Forward Traceability

Requirement

↓

Design

↓

Implementation

↓

Testing

Purpose:

Verify implementation coverage.

1.21 Backward Traceability

Test

↓

Implementation

↓

Design

↓

Requirement

Purpose:

Verify requirement justification.

1.22 Bidirectional Traceability

Required For:

Critical Features

Security Controls

Governance Controls

Audit Controls

Compliance Controls

1.23 Traceability Success Criteria

Success is achieved when:

100% Approved Requirements

↓

Mapped

↓

Implemented

↓

Tested

↓

Verified

1.24 Governance Rules

TRM-GOV-01

Every approved requirement requires traceability.

TRM-GOV-02

Every implementation must map to a requirement.

TRM-GOV-03

Every test must map to a requirement.

TRM-GOV-04

Orphan requirements are prohibited.

TRM-GOV-05

Orphan implementations are prohibited.

TRM-GOV-06

Orphan test cases are prohibited.

1.25 Architectural Constraints

AC-TRM-01

Traceability cannot be optional.

AC-TRM-02

Requirement mappings must remain versioned.

AC-TRM-03

Verification mappings must remain available.

AC-TRM-04

Traceability records must remain auditable.

AC-TRM-05

Traceability records must remain reproducible.

AC-TRM-06

Governance controls cannot be bypassed.

1.26 Section Summary

This section establishes the purpose, scope, objectives, principles, governance model, and operating framework of the TODP Traceability Requirements Matrix.

The TRM provides the authoritative mechanism for linking business requirements, architecture decisions, technical implementation, and validation activities into a single traceable and auditable chain.

TRM Progress

▢ Section 1 — Document Overview & Traceability Scope

▢ Section 2 — Traceability Methodology

TRM V1 — Section 2

Traceability Methodology

Document: Traceability Requirements Matrix (TRM V1)

Project:

TODP (Transparent Opportunity Decision Platform)

Section:

2

Version:

V1

Status:

APPROVED

2.1 Purpose

This section defines the traceability methodology used throughout TODP.

The methodology establishes:

- Requirement mapping standards
- Traceability workflows
- Mapping rules
- Verification procedures
- Governance controls

The methodology ensures that every requirement can be traced from origin to validation.

2.2 Traceability Philosophy

TODP uses:

End-to-End Traceability

Every approved requirement must be traceable across:

Business

Architecture

Design

Implementation

Testing

Operations

2.3 Traceability Direction Types

Forward Traceability

Backward Traceability

Bidirectional Traceability

All critical requirements require bidirectional traceability.

2.4 Forward Traceability Model

Requirement



Architecture



Design



Interface



Implementation



Testing



Verification

Purpose:

Coverage Validation

2.5 Backward Traceability Model

Verification



Test



Implementation



Design



Architecture



Requirement

Purpose:

Justification Validation

2.6 Bidirectional Traceability Model

Requirement



Architecture



Design



Implementation



Testing

Purpose:

Governance Assurance

2.7 Traceability Levels

Level 1

Requirement Traceability

Level 2

Architecture Traceability

Level 3

Design Traceability

Level 4

Implementation Traceability

Level 5

Testing Traceability

Level 6

Operational Traceability

2.8 Requirement Flow

PRD Requirement



ADD Mapping



DDD Mapping



SDD Mapping



IAD Mapping



TDD Mapping



Implementation



Testing

2.9 Traceability Lifecycle

Create Requirement



Approve Requirement



Map Requirement



Implement Requirement



Test Requirement



Verify Requirement



Maintain Requirement

2.10 Mapping Methodology

Each requirement receives:

Unique Identifier



Source Mapping



Target Mapping



Verification Mapping



Status Tracking

2.11 Traceability Granularity

Traceability is maintained at:

Requirement Level

Feature Level

Module Level

API Level

Database Level

Test Case Level

2.12 Requirement Mapping Rules

Every Requirement Must Have:

Source Document

Source Identifier

Target Artifact

Verification Artifact

Status

2.13 Design Mapping Rules

Requirements map to:

Architecture Components

Data Models

Service Components

APIs

Workflows

2.14 Implementation Mapping Rules

Requirements map to:

Modules

Packages

Services

Repositories

Schemas

Infrastructure

2.15 Testing Mapping Rules

Requirements map to:

Unit Tests

Integration Tests

API Tests

Security Tests

Performance Tests

Regression Tests

2.16 Traceability Matrix Entry Structure

Each entry contains:

Requirement ID

Requirement Name

Source

Target

Verification Method

Status

2.17 Source Documents

Approved Sources:

PRD V1

ADD V1

DDD V1

SDD V1

IAD V1

TDD V1

2.18 Target Artifacts

Architecture Components

Database Components

API Components

Service Components

Repository Components

Test Components

2.19 Verification Methods

Review

Inspection

Testing

Validation

Audit

Operational Verification

2.20 Requirement Status Tracking

PROPOSED

APPROVED

IMPLEMENTED

TESTED

VERIFIED

RETIRED

2.21 Coverage Measurement

Coverage Formula:

Mapped Requirements

÷

Approved Requirements

×

100

Target:

100%

2.22 Implementation Coverage

Formula:

Implemented Requirements

÷

Approved Requirements

×

100

Target:

100%

2.23 Testing Coverage

Formula:

Tested Requirements

÷

Implemented Requirements

×

100

Target:

100%

2.24 Verification Coverage

Formula:

Verified Requirements

÷

Tested Requirements

×

100

Target:

100%

2.25 Gap Detection

Gap Categories:

Unmapped Requirement

Unimplemented Requirement

Untested Requirement

Unverified Requirement

2.26 Orphan Detection

Orphan Types:

Orphan Requirement

Orphan Implementation

Orphan API

Orphan Database Object

Orphan Test

2.27 Impact Analysis Method

Requirement Change



Identify Mappings



Identify Dependencies



Identify Affected Components



Identify Affected Tests



Generate Impact Report

2.28 Change Traceability

Every change records:

Change ID

Requirement ID

Affected Artifacts

Approval Reference

Version

2.29 Version Traceability

All mappings include:

Document Version

Requirement Version

Implementation Version

Test Version

2.30 Governance Traceability

Governance controls map to:

Policies

Approvals

Audit Records

Compliance Reports

2.31 Security Traceability

Security requirements map to:

Authentication

Authorization

Encryption

Monitoring

Audit Controls

2.32 Audit Traceability

Audit requirements map to:

Audit APIs

Audit Services

Audit Tables

Audit Tests

2.33 Operational Traceability

Operational requirements map to:

Deployment

Monitoring

Runbooks

Recovery Procedures

Incident Processes

2.34 Traceability Repository

Primary Storage:

Version Controlled Documentation

Secondary Storage:

Implementation References

Test References

Audit References

2.35 Traceability Validation Process

Review Matrix



Validate Mappings



Validate Coverage



Validate Testing



Approve Matrix

2.36 Governance Review Cycle

Review During:

Architecture Reviews

Release Reviews

Security Reviews

Governance Reviews

2.37 Auditability Requirements

Every mapping supports:

Audit Review

Coverage Review

Compliance Review

Historical Review

2.38 Traceability Requirements

Mappings must remain:

Accurate

Versioned

Searchable

Auditable

Reproducible

2.39 Reproducibility Requirements

Support:

Requirement Replay

Design Replay

Implementation Replay

Testing Replay

Verification Replay

2.40 Methodology Success Criteria

Success Requires:

100% Requirement Mapping

100% Implementation Mapping

100% Test Mapping

100% Verification Mapping

2.41 Governance Rules

TRM-METHOD-01

Every requirement must be mapped.

TRM-METHOD-02

Every implementation must have justification.

TRM-METHOD-03

Every test must map to a requirement.

TRM-METHOD-04

Coverage must be measurable.

TRM-METHOD-05

Gaps must be detectable.

TRM-METHOD-06

Traceability must be maintained continuously.

2.42 Architectural Constraints

AC-TRM-METH-01

Manual undocumented mappings are prohibited.

AC-TRM-METH-02

Unmapped requirements are prohibited.

AC-TRM-METH-03

Orphan implementations are prohibited.

AC-TRM-METH-04

Orphan tests are prohibited.

AC-TRM-METH-05

Traceability records cannot be deleted.

AC-TRM-METH-06

Governance controls cannot be bypassed.

2.43 Section Summary

This section defines the traceability methodology used throughout TODP, including mapping workflows, coverage calculations, lifecycle management, gap detection, impact analysis, and governance controls.

The methodology ensures complete end-to-end traceability from business requirements through architecture, design, implementation, testing, and operational validation.

TRM Progress

☐ Section 1 — Document Overview & Traceability Scope

☐ Section 2 — Traceability Methodology

☐ Section 3 — Requirement Identification Standard

TRM V1 — Section 3

Requirement Identification Standard

Document: Traceability Requirements Matrix (TRM V1)

Project:

TODP (Transparent Opportunity Decision Platform)

Section:

3

Version:

V1

Status:

APPROVED

3.1 Purpose

This section defines the Requirement Identification Standard used throughout TODP.

The objective is to ensure that every requirement, design artifact, implementation component, and test asset can be uniquely identified and traced.

The identification standard forms the foundation of the Traceability Requirements Matrix.

3.2 Identification Principles

RID-01

Every requirement receives a unique identifier.

RID-02

Identifiers are immutable.

RID-03

Identifiers remain version traceable.

RID-04

Identifiers support auditability.

RID-05

Identifiers support impact analysis.

RID-06

Identifiers support reproducibility.

3.3 Requirement Identifier Structure

Format:

<DOMAIN>-<CATEGORY>-<NUMBER>

Examples:

PRD-AUTH-001

PRD-OPP-001

PRD-EVID-001

PRD-RANK-001

3.4 Source Document Prefixes

PRD

Product Requirements Document

ADD

Architecture Design Document

DDD

Data Design Document

SDD

Solution Design Document

IAD

Interface Architecture Document

TDD

Technical Design Document

TRM

Traceability Requirements Matrix

3.5 Requirement Domains

AUTH

Authentication

AUTHZ

Authorization

OPP

Opportunity

AUTHORITY

Authority

EVID

Evidence

RETR

Retrieval

RANK

Ranking

REC

Recommendation

GOV

Governance

AUDIT

Audit

SEC

Security

OPS

Operations

3.6 PRD Requirement Standard

Format:

PRD-<DOMAIN>-XXX

Examples:

PRD-AUTH-001

PRD-RETR-005

PRD-RANK-010

3.7 ADD Requirement Standard

Format:

ADD-<DOMAIN>-XXX

Examples:

ADD-SEC-001

ADD-GOV-002

ADD-OPS-003

3.8 DDD Requirement Standard

Format:

DDD-<DOMAIN>-XXX

Examples:

DDD-OPP-001

DDD-EVID-003

DDD-AUDIT-005

3.9 SDD Requirement Standard

Format:

SDD-<DOMAIN>-XXX

Examples:

SDD-RETR-001

SDD-RANK-002

SDD-REC-004

3.10 IAD Requirement Standard

Format:

IAD-<DOMAIN>-XXX

Examples:

IAD-AUTH-001

IAD-REC-005

IAD-GOV-003

3.11 TDD Requirement Standard

Format:

TDD-<DOMAIN>-XXX

Examples:

TDD-SEC-001

TDD-RANK-004

TDD-AUDIT-002

3.12 Database Identifier Standard

Format:

DB-<ENTITY>-XXX

Examples:

DB-USER-001

DB-OPPORTUNITY-001

DB-EVIDENCE-001

3.13 Table Identifier Examples

DB-USERS-001

DB-ROLES-001

DB-PERMISSIONS-001

DB-OPPORTUNITIES-001

DB-EVIDENCES-001

3.14 API Identifier Standard

Format:

API-<DOMAIN>-XXX

Examples:

API-AUTH-001

API-OPP-001

API-RANK-001

API-REC-001

3.15 Endpoint Identifier Examples

API-AUTH-LOGIN-001

API-AUTH-REFRESH-002

API-OPP-CREATE-001

API-REC-GENERATE-001

3.16 Service Identifier Standard

Format:

SRV-<DOMAIN>-XXX

Examples:

SRV-AUTH-001

SRV-EVID-001

SRV-RANK-001

3.17 Repository Identifier Standard

Format:

REP-<DOMAIN>-XXX

Examples:

REP-OPP-001

REP-AUTHORITY-001

REP-AUDIT-001

3.18 Event Identifier Standard

Format:

EVT-<DOMAIN>-XXX

Examples:

EVT-AUTH-001

EVT-RANK-001

EVT-REC-001

3.19 Test Identifier Standard

Format:

TEST-<TYPE>-XXX

Examples:

TEST-UNIT-001

TEST-INT-001

TEST-API-001

TEST-SEC-001

TEST-E2E-001

3.20 Unit Test Standard

Format:

TEST-UNIT-<DOMAIN>-XXX

Examples:

TEST-UNIT-AUTH-001

TEST-UNIT-RANK-002

3.21 Integration Test Standard

Format:

TEST-INT-<DOMAIN>-XXX

Examples:

TEST-INT-EVID-001

TEST-INT-REC-001

3.22 API Test Standard

Format:

TEST-API-<DOMAIN>-XXX

Examples:

TEST-API-AUTH-001

TEST-API-OPP-002

3.23 Security Test Standard

Format:

TEST-SEC-<DOMAIN>-XXX

Examples:

TEST-SEC-AUTH-001

TEST-SEC-GOV-002

3.24 Performance Test Standard

Format:

TEST-PERF-<DOMAIN>-XXX

Examples:

TEST-PERF-RETR-001

TEST-PERF-RANK-001

3.25 Requirement Relationship Identifier

Format:

REL-XXX

Purpose:

Relationship Tracking

Examples:

REL-001

REL-002

REL-003

3.26 Change Identifier Standard

Format:

CHG-YYYY-XXX

Examples:

CHG-2026-001

CHG-2026-002

3.27 Version Identifier Standard

Format:

V<MAJOR>.<MINOR>

Examples:

V1.0

V1.1

V2.0

3.28 Traceability Record Identifier

Format:

TRACE-XXX

Examples:

TRACE-001

TRACE-002

TRACE-003

3.29 Audit Identifier Standard

Format:

AUD-XXX

Examples:

AUD-001

AUD-002

AUD-003

3.30 Incident Identifier Standard

Format:

INC-YYYY-XXX

Examples:

INC-2026-001

INC-2026-002

3.31 Governance Identifier Standard

Format:

GOV-XXX

Examples:

GOV-001

GOV-002

GOV-003

3.32 Identifier Uniqueness Rules

Identifiers Must Be:

Unique

Immutable

Non-Reusable

Version Traceable

Auditable

3.33 Identifier Assignment Rules

Assigned Once

Never Recycled

Never Reassigned

3.34 Identifier Retirement Rules

Retired Identifiers:

Remain Traceable

Remain Auditable

Remain Searchable

3.35 Traceability Mapping Format

Example:

PRD-RANK-001



ADD-RANK-002



DDD-RANK-001



SDD-RANK-003



IAD-RANK-001



TDD-RANK-004



API-RANK-001



TEST-API-RANK-001

3.36 Coverage Identification Rules

Every Mapping Must Include:

Source ID

Target ID

Status

Version

Verification Reference

3.37 Auditability Requirements

Identifiers Support:

Requirement Audits

Implementation Audits

Testing Audits

Governance Audits

3.38 Traceability Requirements

Identifiers Support:

Forward Tracing

Backward Tracing

Impact Analysis

Dependency Analysis

3.39 Reproducibility Requirements

Identifiers Support:

Requirement Replay

Version Replay

Implementation Replay

Test Replay

3.40 Governance Rules

RID-GOV-01

Every requirement requires an identifier.

RID-GOV-02

Identifiers are immutable.

RID-GOV-03

Identifiers cannot be reused.

RID-GOV-04

Identifiers must be traceable.

RID-GOV-05

Identifiers must be auditable.

RID-GOV-06

Identifier standards must remain consistent.

3.41 Architectural Constraints

AC-RID-01

Duplicate identifiers are prohibited.

AC-RID-02

Identifier reassignment is prohibited.

AC-RID-03

Identifier deletion is prohibited.

AC-RID-04

Traceability without identifiers is prohibited.

AC-RID-05

Identifier governance cannot be bypassed.

AC-RID-06

Version traceability is mandatory.

3.42 Section Summary

This section defines the requirement identification standards, naming conventions, mapping structures, versioning rules, and governance controls used throughout TODP.

The identification framework ensures that every requirement, design artifact, implementation component, test asset, and governance artifact can be uniquely identified, traced, audited, and reproduced throughout the lifecycle of the platform.

TRM Progress

📄 Section 1 — Document Overview & Traceability Scope

📄 Section 2 — Traceability Methodology

📄 Section 3 — Requirement Identification Standard

📄 Section 4 — PRD → ADD Mapping

TRM V1 — Section 4

PRD → ADD Mapping

Document: Traceability Requirements Matrix (TRM V1)

Project:

TODP (Transparent Opportunity Decision Platform)

Section:

4

Version:

V1

Status:

APPROVED

4.1 Purpose

This section establishes traceability between:

PRD V1

↓

ADD V1

The objective is to verify that every approved business requirement has corresponding architectural coverage.

4.2 Mapping Principles

Business Requirement

↓

Architecture Decision

Every PRD requirement must be supported by at least one ADD architectural construct.

4.3 Authentication Domain Mapping

PRD Requirement	ADD Mapping	Coverage Status
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PRD-AUTH-001 User Authentication	ADD-AUTH-001 Identity Architecture	COVERED
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PRD-AUTH-002 Session Management	ADD-AUTH-002 Session Architecture	COVERED
---------------------------------	-----------------------------------	---------

PRD-AUTH-003 Token Management	ADD-AUTH-003 Token Architecture	COVERED
-------------------------------	---------------------------------	---------

| PRD-AUTH-004 Password Management | ADD-AUTH-004 Credential Architecture | COVERED |

4.4 Authorization Domain Mapping

| PRD Requirement | ADD Mapping | Coverage Status |

|-----|-----|-----|

| PRD-AUTHZ-001 Role Based Access Control | ADD-AUTHZ-001 Authorization Architecture |
COVERED |

| PRD-AUTHZ-002 Permission Management | ADD-AUTHZ-002 Permission Architecture | COVERED |

| PRD-AUTHZ-003 Governance Access Controls | ADD-GOV-001 Governance Security Model |
COVERED |

4.5 Opportunity Domain Mapping

| PRD Requirement | ADD Mapping | Coverage Status |

|-----|-----|-----|

| PRD-OPP-001 Opportunity Creation | ADD-OPP-001 Opportunity Architecture | COVERED |

| PRD-OPP-002 Opportunity Management | ADD-OPP-002 Opportunity Lifecycle Architecture |
COVERED |

| PRD-OPP-003 Opportunity Search | ADD-RETR-001 Retrieval Architecture | COVERED |

| PRD-OPP-004 Opportunity Versioning | ADD-OPP-003 Version Management Architecture |
COVERED |

4.6 Authority Domain Mapping

PRD Requirement	ADD Mapping	Coverage Status
-----	-----	-----
PRD-AUTHORITY-001 Authority Creation	ADD-AUTHORITY-001 Authority Architecture	COVERED
PRD-AUTHORITY-002 Authority Verification	ADD-AUTHORITY-002 Verification Architecture	COVERED
PRD-AUTHORITY-003 Confidence Scoring	ADD-AUTHORITY-003 Confidence Architecture	COVERED

4.7 Evidence Domain Mapping

PRD Requirement	ADD Mapping	Coverage Status
-----	-----	-----
PRD-EVID-001 Evidence Management	ADD-EVID-001 Evidence Architecture	COVERED
PRD-EVID-002 Evidence Validation	ADD-EVID-002 Validation Architecture	COVERED
PRD-EVID-003 Evidence Versioning	ADD-EVID-003 Version Architecture	COVERED
PRD-EVID-004 Evidence Lineage	ADD-EVID-004 Lineage Architecture	COVERED

4.8 Retrieval Domain Mapping

PRD Requirement	ADD Mapping	Coverage Status
-----	-----	-----
PRD-RETR-001 Search Capability	ADD-RETR-001 Retrieval Architecture	COVERED

PRD-RETR-002 Keyword Retrieval	ADD-RETR-002 BM25 Architecture	COVERED
PRD-RETR-003 Semantic Retrieval	ADD-RETR-003 Vector Architecture	COVERED
PRD-RETR-004 Hybrid Retrieval	ADD-RETR-004 Hybrid Search Architecture	COVERED
PRD-RETR-005 Retrieval Replay	ADD-RETR-005 Replay Architecture	COVERED

4.9 Ranking Domain Mapping

PRD Requirement	ADD Mapping	Coverage Status
-----	-----	-----
PRD-RANK-001 Ranking Execution	ADD-RANK-001 Ranking Architecture	COVERED
PRD-RANK-002 Total Cost Formula	ADD-RANK-002 Deterministic Ranking Architecture	COVERED
PRD-RANK-003 Ranking Replay	ADD-RANK-003 Replay Architecture	COVERED
PRD-RANK-004 Formula Versioning	ADD-RANK-004 Version Architecture	COVERED

4.10 Recommendation Domain Mapping

PRD Requirement	ADD Mapping	Coverage Status
-----	-----	-----
PRD-REC-001 Recommendation Generation	ADD-REC-001 Recommendation Architecture	COVERED
PRD-REC-002 Recommendation Explanation	ADD-REC-002 Explainability Architecture	COVERED
PRD-REC-003 Recommendation Traceability	ADD-REC-003 Traceability Architecture	COVERED
PRD-REC-004 Recommendation Reproducibility	ADD-REC-004 Replay Architecture	COVERED

4.11 Governance Domain Mapping

PRD Requirement	ADD Mapping	Coverage Status
-----	-----	-----
PRD-GOV-001 Governance Framework	ADD-GOV-001 Governance Architecture	COVERED
PRD-GOV-002 Policy Validation	ADD-GOV-002 Policy Architecture	COVERED
PRD-GOV-003 Approval Workflow	ADD-GOV-003 Approval Architecture	COVERED
PRD-GOV-004 Compliance Reporting	ADD-GOV-004 Compliance Architecture	COVERED

4.12 Audit Domain Mapping

PRD Requirement	ADD Mapping	Coverage Status
-----	-----	-----
PRD-AUDIT-001 Auditability	ADD-AUDIT-001 Audit Architecture	COVERED
PRD-AUDIT-002 Replay Support	ADD-AUDIT-002 Replay Architecture	COVERED
PRD-AUDIT-003 Snapshot Support	ADD-AUDIT-003 Snapshot Architecture	COVERED
PRD-AUDIT-004 Correlation Tracking	ADD-AUDIT-004 Correlation Architecture	COVERED

4.13 Security Domain Mapping

PRD Requirement	ADD Mapping	Coverage Status
-----	-----	-----
PRD-SEC-001 Authentication Security	ADD-SEC-001 Identity Security Architecture	COVERED
PRD-SEC-002 Authorization Security	ADD-SEC-002 Access Control Architecture	COVERED
PRD-SEC-003 Encryption	ADD-SEC-003 Encryption Architecture	COVERED
PRD-SEC-004 Secret Management	ADD-SEC-004 Secret Management Architecture	COVERED

4.14 Observability Domain Mapping

PRD Requirement	ADD Mapping	Coverage Status
-----	-----	-----
PRD-OPS-001 Logging	ADD-OBS-001 Logging Architecture	COVERED
PRD-OPS-002 Monitoring	ADD-OBS-002 Monitoring Architecture	COVERED
PRD-OPS-003 Tracing	ADD-OBS-003 Tracing Architecture	COVERED
PRD-OPS-004 Alerting	ADD-OBS-004 Alerting Architecture	COVERED

4.15 Deployment Domain Mapping

PRD Requirement	ADD Mapping	Coverage Status
-----	-----	-----
PRD-DEP-001 Deployment Architecture	ADD-DEP-001 Deployment Architecture	COVERED
PRD-DEP-002 Environment Isolation	ADD-DEP-002 Environment Architecture	COVERED
PRD-DEP-003 Scalability	ADD-DEP-003 Scaling Architecture	COVERED

4.16 Coverage Summary

Business Domains Covered:

Authentication

Authorization

Opportunity

Authority

Evidence

Retrieval

Ranking

Recommendation

Governance

Audit

Security

Operations

Deployment

Coverage Status:

100% Covered

4.17 Mapping Verification Rules

Verification Requires:

PRD Requirement Exists

↓

ADD Mapping Exists

↓

Architecture Coverage Confirmed



Governance Approval Confirmed

4.18 Gap Detection

Current Status:

No Unmapped PRD Requirements Identified

Coverage Gap Count:

0

4.19 Forward Traceability

PRD Requirement



ADD Architecture



DDD Design



SDD Solution



IAD Interface



TDD Technical Design



4.20 Backward Traceability

Architecture Component



Mapped Requirement



Business Justification

4.21 Auditability Requirements

All mappings support:

Architecture Audits

Requirement Audits

Coverage Reviews

Governance Reviews

4.22 Traceability Requirements

Mappings remain:

Version Controlled

Searchable

Auditable

Reproducible

4.23 Reproducibility Requirements

Support:

Requirement Replay

Architecture Replay

Coverage Replay

Governance Replay

4.24 Governance Rules

TRM-ADD-01

Every PRD requirement requires architectural coverage.

TRM-ADD-02

Unmapped business requirements are prohibited.

TRM-ADD-03

Architecture decisions require business justification.

TRM-ADD-04

Coverage must be measurable.

TRM-ADD-05

Mappings must remain versioned.

TRM-ADD-06

Coverage reviews are mandatory.

4.25 Architectural Constraints

AC-TRM-ADD-01

Business requirements cannot bypass architecture.

AC-TRM-ADD-02

Architecture without requirements is prohibited.

AC-TRM-ADD-03

Unjustified architecture is prohibited.

AC-TRM-ADD-04

Coverage gaps require remediation.

AC-TRM-ADD-05

Mappings cannot be deleted.

AC-TRM-ADD-06

Governance controls cannot be bypassed.

4.26 Section Summary

This section establishes traceability between PRD V1 and ADD V1, ensuring that all approved business requirements have corresponding architectural coverage.

The mapping confirms that TODP architecture fully supports the approved business objectives and provides complete architectural coverage across all functional, operational, governance, security, and audit domains.

TRM Progress

▣ Section 1 — Document Overview & Traceability Scope

▣ Section 2 — Traceability Methodology

▣ Section 3 — Requirement Identification Standard

▣ Section 4 — PRD → ADD Mapping

▣ Section 5 — PRD → DDD Mapping

TRM V1 — Section 5

PRD → DDD Mapping

Document: Traceability Requirements Matrix (TRM V1)

Project:

TODP (Transparent Opportunity Decision Platform)

Section:

5

Version:

V1

Status:

APPROVED

5.1 Purpose

This section establishes traceability between:

PRD V1

↓

DDD V1

The objective is to verify that every approved business requirement has corresponding data model, entity, schema, and persistence coverage.

5.2 Mapping Principles

Business Requirement

↓

Data Requirement

Every PRD requirement must map to one or more DDD entities, relationships, schemas, constraints, or persistence structures.

5.3 Authentication Domain Mapping

PRD Requirement	DDD Mapping	Coverage Status
-----	-----	-----
PRD-AUTH-001 User Authentication	DDD-AUTH-001 User Entity	COVERED
PRD-AUTH-002 Session Management	DDD-AUTH-002 Session Entity	COVERED
PRD-AUTH-003 Token Management	DDD-AUTH-003 Token Persistence Model	COVERED
PRD-AUTH-004 Password Management	DDD-AUTH-004 Credential Data Model	COVERED

5.4 Authorization Domain Mapping

PRD Requirement	DDD Mapping	Coverage Status
-----	-----	-----
PRD-AUTHZ-001 RBAC	DDD-AUTHZ-001 Role Entity	COVERED
PRD-AUTHZ-002 Permission Management	DDD-AUTHZ-002 Permission Entity	COVERED
PRD-AUTHZ-003 Access Controls	DDD-AUTHZ-003 Role Permission Mapping	COVERED

5.5 Opportunity Domain Mapping

PRD Requirement	DDD Mapping	Coverage Status
-----	-----	-----
PRD-OPP-001 Opportunity Creation	DDD-OPP-001 Opportunity Entity	COVERED
PRD-OPP-002 Opportunity Management	DDD-OPP-002 Opportunity Lifecycle Model	COVERED
PRD-OPP-003 Opportunity Search	DDD-OPP-003 Searchable Opportunity Schema	COVERED
PRD-OPP-004 Opportunity Versioning	DDD-OPP-004 Opportunity Version Model	COVERED

5.6 Authority Domain Mapping

PRD Requirement	DDD Mapping	Coverage Status
-----	-----	-----
PRD-AUTHORITY-001 Authority Creation	DDD-AUTHORITY-001 Authority Entity	COVERED
PRD-AUTHORITY-002 Authority Verification	DDD-AUTHORITY-002 Verification Model	COVERED
PRD-AUTHORITY-003 Confidence Scoring	DDD-AUTHORITY-003 Confidence Data Model	COVERED

5.7 Evidence Domain Mapping

PRD Requirement	DDD Mapping	Coverage Status
-----	-----	-----
PRD-EVID-001 Evidence Management	DDD-EVID-001 Evidence Entity	COVERED
PRD-EVID-002 Evidence Validation	DDD-EVID-002 Validation Model	COVERED
PRD-EVID-003 Evidence Versioning	DDD-EVID-003 Evidence Version Entity	COVERED
PRD-EVID-004 Evidence Lineage	DDD-EVID-004 Lineage Relationship Model	COVERED

5.8 Retrieval Domain Mapping

PRD Requirement	DDD Mapping	Coverage Status
-----	-----	-----
PRD-RETR-001 Search Capability	DDD-RETR-001 Search Entity	COVERED
PRD-RETR-002 Keyword Retrieval	DDD-RETR-002 Search Metadata Model	COVERED
PRD-RETR-003 Semantic Retrieval	DDD-RETR-003 Embedding Entity	COVERED
PRD-RETR-004 Hybrid Retrieval	DDD-RETR-004 Hybrid Search Result Model	COVERED
PRD-RETR-005 Search Replay	DDD-RETR-005 Search History Entity	COVERED

5.9 Ranking Domain Mapping

PRD Requirement	DDD Mapping	Coverage Status
-----	-----	-----
PRD-RANK-001 Ranking Execution	DDD-RANK-001 Ranking Execution Entity	COVERED
PRD-RANK-002 Total Cost Formula	DDD-RANK-002 Ranking Result Schema	COVERED

| PRD-RANK-003 Ranking Replay | DDD-RANK-003 Ranking History Model | COVERED |
| PRD-RANK-004 Formula Versioning | DDD-RANK-004 Formula Version Entity | COVERED |

5.10 Recommendation Domain Mapping

PRD Requirement	DDD Mapping	Coverage Status
PRD-REC-001 Recommendation Generation	DDD-REC-001 Recommendation Entity	COVERED
PRD-REC-002 Recommendation Explanation	DDD-REC-002 Explanation Model	COVERED
PRD-REC-003 Recommendation Traceability	DDD-REC-003 Recommendation Link Model	
COVERED		
PRD-REC-004 Recommendation Replay	DDD-REC-004 Recommendation History Entity	COVERED

5.11 Governance Domain Mapping

PRD Requirement	DDD Mapping	Coverage Status
PRD-GOV-001 Governance Framework	DDD-GOV-001 Governance Entity	COVERED
PRD-GOV-002 Policy Validation	DDD-GOV-002 Policy Entity	COVERED
PRD-GOV-003 Approval Workflow	DDD-GOV-003 Approval Entity	COVERED
PRD-GOV-004 Compliance Reporting	DDD-GOV-004 Compliance Entity	COVERED

5.12 Audit Domain Mapping

PRD Requirement	DDD Mapping	Coverage Status
-----	-----	-----
PRD-AUDIT-001 Auditability	DDD-AUDIT-001 Audit Record Entity	COVERED
PRD-AUDIT-002 Replay Support	DDD-AUDIT-002 Replay Model	COVERED
PRD-AUDIT-003 Snapshot Support	DDD-AUDIT-003 Snapshot Entity	COVERED
PRD-AUDIT-004 Correlation Tracking	DDD-AUDIT-004 Correlation Entity	COVERED

5.13 Security Domain Mapping

PRD Requirement	DDD Mapping	Coverage Status
-----	-----	-----
PRD-SEC-001 Identity Security	DDD-SEC-001 Identity Data Model	COVERED
PRD-SEC-002 Access Security	DDD-SEC-002 Access Control Model	COVERED
PRD-SEC-003 Encryption Support	DDD-SEC-003 Protected Data Structures	COVERED
PRD-SEC-004 Secret Management	DDD-SEC-004 Secret Reference Model	COVERED

5.14 Operational Domain Mapping

PRD Requirement	DDD Mapping	Coverage Status
-----	-----	-----

PRD-OPS-001 Logging	DDD-OPS-001 Log Metadata Model	COVERED
PRD-OPS-002 Monitoring	DDD-OPS-002 Monitoring Data Model	COVERED
PRD-OPS-003 Tracing	DDD-OPS-003 Trace Entity	COVERED
PRD-OPS-004 Alerting	DDD-OPS-004 Alert Entity	COVERED

5.15 Deployment Domain Mapping

PRD Requirement	DDD Mapping	Coverage Status
-----	-----	-----
PRD-DEP-001 Deployment Architecture	DDD-DEP-001 Deployment Metadata Model	COVERED
PRD-DEP-002 Environment Management	DDD-DEP-002 Environment Entity	COVERED
PRD-DEP-003 Scalability	DDD-DEP-003 Capacity Metadata Model	COVERED
PRD-DEP-004 Disaster Recovery	DDD-DEP-004 Backup & Recovery Entity	COVERED

5.16 Core Entity Coverage

Major Entities Covered:

Users

Roles

Permissions

Sessions

Opportunities

Authorities

Evidence

Searches

Embeddings

Rankings

Recommendations

Governance

Audit Records

Snapshots

Traces

Coverage Status:

100%

5.17 Relationship Coverage

Relationships Covered:

User ↔ Role

Role ↔ Permission

Opportunity ↔ Evidence

Authority ↔ Evidence

Search ↔ Ranking

Ranking ↔ Recommendation

Governance ↔ Approval

Audit ↔ Trace

Coverage Status:

100%

5.18 Persistence Coverage

Covered Storage Domains:

Authentication

Authorization

Opportunity

Authority

Evidence

Retrieval

Ranking

Recommendation

Governance

Audit

Operations

Coverage Status:

100%

5.19 Gap Detection

Current Assessment:

No Missing Data Coverage Identified

Unmapped Requirements:

0

5.20 Forward Traceability

PRD Requirement



DDD Entity



Database Table



Repository



Service



API

5.21 Backward Traceability

Database Entity



DDD Model



Mapped Requirement



Business Justification

5.22 Verification Method

Verification Steps:

Requirement Review



Entity Review



Relationship Review



Coverage Validation



Governance Approval

5.23 Auditability Requirements

Mappings Support:

Data Audits

Schema Audits

Coverage Audits

Governance Reviews

5.24 Traceability Requirements

Mappings Remain:

Version Controlled

Auditable

Searchable

Reproducible

5.25 Reproducibility Requirements

Support:

Schema Replay

Entity Replay

Version Replay

Coverage Replay

5.26 Governance Rules

TRM-DDD-01

Every PRD requirement requires data coverage.

TRM-DDD-02

Every persistent feature requires entity coverage.

TRM-DDD-03

Every relationship requires documentation.

TRM-DDD-04

Coverage must be measurable.

TRM-DDD-05

Mappings must remain version controlled.

TRM-DDD-06

Gap analysis is mandatory.

5.27 Architectural Constraints

AC-TRM-DDD-01

Business requirements cannot bypass data design.

AC-TRM-DDD-02

Persistent data without requirements is prohibited.

AC-TRM-DDD-03

Unmapped entities are prohibited.

AC-TRM-DDD-04

Coverage gaps require remediation.

AC-TRM-DDD-05

Mappings cannot be deleted.

AC-TRM-DDD-06

Governance controls cannot be bypassed.

5.28 Section Summary

This section establishes traceability between PRD V1 and DDD V1, ensuring that every approved business requirement has corresponding entity, relationship, schema, and persistence coverage.

The mapping confirms complete data architecture support for all TODP functional, operational, governance, security, and audit requirements.

TRM Progress

📄 Section 1 — Document Overview & Traceability Scope

📄 Section 2 — Traceability Methodology

📄 Section 3 — Requirement Identification Standard

📄 Section 4 — PRD → ADD Mapping

📄 Section 5 — PRD → DDD Mapping

📄 Section 6 — PRD → SDD Mapping

TRM V1 — Section 6

PRD → SDD Mapping

Document: Traceability Requirements Matrix (TRM V1)

Project:

TODP (Transparent Opportunity Decision Platform)

Section:

6

Version:

V1

Status:

APPROVED

6.1 Purpose

This section establishes traceability between:

PRD V1

↓

SDD V1

The objective is to verify that every approved business requirement has corresponding solution design coverage.

The SDD provides the workflow, component interaction, service orchestration, and solution-level realization of business requirements.

6.2 Mapping Principles

Business Requirement



Solution Design

Every approved PRD requirement must be represented by one or more solution workflows, services, orchestration flows, or operational processes.

6.3 Authentication Domain Mapping

PRD Requirement	SDD Mapping	Coverage Status
-----	-----	-----
PRD-AUTH-001 User Authentication	SDD-AUTH-001 Authentication Workflow	COVERED
PRD-AUTH-002 Session Management	SDD-AUTH-002 Session Lifecycle Workflow	COVERED
PRD-AUTH-003 Token Management	SDD-AUTH-003 Token Workflow	COVERED

| PRD-AUTH-004 Password Management | SDD-AUTH-004 Credential Management Workflow |
COVERED |

6.4 Authorization Domain Mapping

| PRD Requirement | SDD Mapping | Coverage Status |

|-----|-----|-----|

| PRD-AUTHZ-001 RBAC | SDD-AUTHZ-001 Authorization Workflow | COVERED |

| PRD-AUTHZ-002 Permission Management | SDD-AUTHZ-002 Permission Evaluation Flow |
COVERED |

| PRD-AUTHZ-003 Governance Access Controls | SDD-GOV-001 Governance Access Workflow |
COVERED |

6.5 Opportunity Domain Mapping

| PRD Requirement | SDD Mapping | Coverage Status |

|-----|-----|-----|

| PRD-OPP-001 Opportunity Creation | SDD-OPP-001 Opportunity Creation Workflow | COVERED |

| PRD-OPP-002 Opportunity Management | SDD-OPP-002 Opportunity Lifecycle Workflow |
COVERED |

| PRD-OPP-003 Opportunity Search | SDD-OPP-003 Opportunity Search Workflow | COVERED |

| PRD-OPP-004 Opportunity Versioning | SDD-OPP-004 Version Management Workflow | COVERED
|

6.6 Authority Domain Mapping

PRD Requirement SDD Mapping Coverage Status
----- ----- -----
PRD-AUTHORITY-001 Authority Creation SDD-AUTHORITY-001 Authority Workflow COVERED
PRD-AUTHORITY-002 Authority Verification SDD-AUTHORITY-002 Verification Workflow COVERED
PRD-AUTHORITY-003 Confidence Scoring SDD-AUTHORITY-003 Confidence Evaluation Flow COVERED

6.7 Evidence Domain Mapping

PRD Requirement SDD Mapping Coverage Status
----- ----- -----
PRD-EVID-001 Evidence Management SDD-EVID-001 Evidence Workflow COVERED
PRD-EVID-002 Evidence Validation SDD-EVID-002 Validation Workflow COVERED
PRD-EVID-003 Evidence Versioning SDD-EVID-003 Version Workflow COVERED
PRD-EVID-004 Evidence Lineage SDD-EVID-004 Lineage Workflow COVERED

6.8 Retrieval Domain Mapping

PRD Requirement SDD Mapping Coverage Status
----- ----- -----

PRD-RETR-001 Search Capability	SDD-RETR-001 Search Workflow	COVERED
PRD-RETR-002 Keyword Retrieval	SDD-RETR-002 BM25 Retrieval Workflow	COVERED
PRD-RETR-003 Semantic Retrieval	SDD-RETR-003 Vector Retrieval Workflow	COVERED
PRD-RETR-004 Hybrid Retrieval	SDD-RETR-004 Hybrid Retrieval Workflow	COVERED
PRD-RETR-005 Search Replay	SDD-RETR-005 Search Replay Workflow	COVERED

6.9 Ranking Domain Mapping

PRD Requirement	SDD Mapping	Coverage Status
-----	-----	-----
PRD-RANK-001 Ranking Execution	SDD-RANK-001 Ranking Workflow	COVERED
PRD-RANK-002 Total Cost Formula	SDD-RANK-002 Formula Evaluation Workflow	COVERED
PRD-RANK-003 Ranking Replay	SDD-RANK-003 Replay Workflow	COVERED
PRD-RANK-004 Formula Versioning	SDD-RANK-004 Formula Lifecycle Workflow	COVERED

6.10 Recommendation Domain Mapping

PRD Requirement	SDD Mapping	Coverage Status
-----	-----	-----
PRD-REC-001 Recommendation Generation	SDD-REC-001 Recommendation Workflow	COVERED
PRD-REC-002 Recommendation Explanation	SDD-REC-002 Explanation Workflow	COVERED
PRD-REC-003 Recommendation Traceability	SDD-REC-003 Traceability Workflow	COVERED
PRD-REC-004 Recommendation Replay	SDD-REC-004 Replay Workflow	COVERED

6.11 Governance Domain Mapping

PRD Requirement SDD Mapping Coverage Status
----- ----- -----
PRD-GOV-001 Governance Framework SDD-GOV-001 Governance Workflow COVERED
PRD-GOV-002 Policy Validation SDD-GOV-002 Policy Evaluation Workflow COVERED
PRD-GOV-003 Approval Workflow SDD-GOV-003 Approval Process Flow COVERED
PRD-GOV-004 Compliance Reporting SDD-GOV-004 Compliance Workflow COVERED

6.12 Audit Domain Mapping

PRD Requirement SDD Mapping Coverage Status
----- ----- -----
PRD-AUDIT-001 Auditability SDD-AUDIT-001 Audit Workflow COVERED
PRD-AUDIT-002 Replay Support SDD-AUDIT-002 Replay Workflow COVERED
PRD-AUDIT-003 Snapshot Support SDD-AUDIT-003 Snapshot Workflow COVERED
PRD-AUDIT-004 Correlation Tracking SDD-AUDIT-004 Correlation Workflow COVERED

6.13 Security Domain Mapping

PRD Requirement	SDD Mapping	Coverage Status
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-----	-----	-----
-------	-------	-------

PRD-SEC-001 Authentication Security	SDD-SEC-001 Security Workflow	COVERED
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PRD-SEC-002 Authorization Security	SDD-SEC-002 Access Validation Workflow	COVERED
------------------------------------	--	---------

PRD-SEC-003 Encryption	SDD-SEC-003 Data Protection Workflow	COVERED
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PRD-SEC-004 Secret Management	SDD-SEC-004 Secret Handling Workflow	COVERED
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6.14 Operational Domain Mapping

PRD Requirement	SDD Mapping	Coverage Status
-----------------	-------------	-----------------

-----	-----	-----
-------	-------	-------

PRD-OPS-001 Logging	SDD-OPS-001 Logging Workflow	COVERED
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PRD-OPS-002 Monitoring	SDD-OPS-002 Monitoring Workflow	COVERED
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PRD-OPS-003 Tracing	SDD-OPS-003 Trace Workflow	COVERED
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PRD-OPS-004 Alerting	SDD-OPS-004 Alert Management Workflow	COVERED
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6.15 Deployment Domain Mapping

PRD Requirement	SDD Mapping	Coverage Status
-----------------	-------------	-----------------

-----	-----	-----
-------	-------	-------

PRD-DEP-001 Deployment Architecture	SDD-DEP-001 Deployment Workflow	COVERED
-------------------------------------	---------------------------------	---------

PRD-DEP-002 Environment Management	SDD-DEP-002 Environment Workflow	COVERED
------------------------------------	----------------------------------	---------

PRD-DEP-003 Scalability	SDD-DEP-003 Scaling Workflow	COVERED
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6.16 Workflow Coverage Summary

Covered Workflows:

Authentication

Authorization

Opportunity

Authority

Evidence

Retrieval

Ranking

Recommendation

Governance

Audit

Security

Operations

Deployment

Coverage Status:

100%

6.17 Service Interaction Coverage

Covered Interactions:

API ↔ Service

Service ↔ Repository

Repository ↔ Database

Service ↔ Audit

Service ↔ Governance

Service ↔ Observability

Coverage Status:

100%

6.18 Process Coverage

Covered Processes:

Validation

Approval

Retrieval

Ranking

Recommendation

Audit

Replay

Recovery

Coverage Status:

100%

6.19 Gap Detection

Current Assessment:

No Missing Solution Coverage Identified

Coverage Gaps:

0

6.20 Forward Traceability

PRD Requirement



SDD Workflow



IAD Contract



TDD Component



Implementation

6.21 Backward Traceability

Workflow



Mapped Requirement



Business Objective

6.22 Verification Method

Verification Steps:

Requirement Review



Workflow Review



Coverage Validation



Governance Approval

6.23 Auditability Requirements

Mappings Support:

Workflow Audits

Coverage Reviews

Governance Reviews

Operational Reviews

6.24 Traceability Requirements

Mappings Remain:

Version Controlled

Auditable

Searchable

Reproducible

6.25 Reproducibility Requirements

Support:

Workflow Replay

Process Replay

Coverage Replay

Validation Replay

6.26 Governance Rules

TRM-SDD-01

Every business requirement requires workflow coverage.

TRM-SDD-02

Every workflow requires business justification.

TRM-SDD-03

Coverage must be measurable.

TRM-SDD-04

Solution gaps require remediation.

TRM-SDD-05

Mappings must remain version controlled.

TRM-SDD-06

Coverage validation is mandatory.

6.27 Architectural Constraints

AC-TRM-SDD-01

Business requirements cannot bypass solution design.

AC-TRM-SDD-02

Unmapped workflows are prohibited.

AC-TRM-SDD-03

Unjustified workflows are prohibited.

AC-TRM-SDD-04

Coverage gaps require remediation.

AC-TRM-SDD-05

Mappings cannot be deleted.

AC-TRM-SDD-06

Governance controls cannot be bypassed.

6.28 Section Summary

This section establishes traceability between PRD V1 and SDD V1, ensuring that every approved business requirement has corresponding workflow, orchestration, process, and solution design coverage.

The mapping confirms complete solution-level realization of TODP business requirements across all functional, operational, governance, security, and audit domains.

TRM Progress

☐ Section 1 — Document Overview & Traceability Scope

☐ Section 2 — Traceability Methodology

☐ Section 3 — Requirement Identification Standard

☐ Section 4 — PRD → ADD Mapping

☐ Section 5 — PRD → DDD Mapping

☐ Section 6 — PRD → SDD Mapping

☐ Section 7 — PRD → IAD Mapping

TRM V1 — Section 7

PRD → IAD Mapping

Document: Traceability Requirements Matrix (TRM V1)

Project:

TODP (Transparent Opportunity Decision Platform)

Section:

7

Version:

V1

Status:

APPROVED

7.1 Purpose

This section establishes traceability between:

PRD V1

↓

IAD V1

The objective is to verify that every approved business requirement has corresponding interface, API, contract, request-response, and integration coverage.

The Interface Architecture Document defines how external and internal consumers interact with TODP capabilities.

7.2 Mapping Principles

Business Requirement

↓

Interface Contract

Every approved PRD requirement must map to one or more interface contracts defined within IAD V1.

7.3 Authentication Domain Mapping

PRD Requirement IAD Mapping Coverage Status
----- ----- -----
PRD-AUTH-001 User Authentication IAD-AUTH-001 Login API Contract COVERED
PRD-AUTH-002 Session Management IAD-AUTH-002 Session Contract COVERED
PRD-AUTH-003 Token Management IAD-AUTH-003 Token Refresh Contract COVERED
PRD-AUTH-004 Password Management IAD-AUTH-004 Credential Contract COVERED

7.4 Authorization Domain Mapping

PRD Requirement IAD Mapping Coverage Status
----- ----- -----
PRD-AUTHZ-001 RBAC IAD-AUTHZ-001 Role Contract COVERED
PRD-AUTHZ-002 Permission Management IAD-AUTHZ-002 Permission Contract COVERED
PRD-AUTHZ-003 Access Controls IAD-AUTHZ-003 Authorization Contract COVERED

7.5 Opportunity Domain Mapping

PRD Requirement IAD Mapping Coverage Status
----- ----- -----
PRD-OPP-001 Opportunity Creation IAD-OPP-001 Create Opportunity API COVERED
PRD-OPP-002 Opportunity Management IAD-OPP-002 Opportunity CRUD APIs COVERED
PRD-OPP-003 Opportunity Search IAD-OPP-003 Search APIs COVERED
PRD-OPP-004 Opportunity Versioning IAD-OPP-004 Version APIs COVERED

7.6 Authority Domain Mapping

PRD Requirement IAD Mapping Coverage Status
----- ----- -----
PRD-AUTHORITY-001 Authority Creation IAD-AUTHORITY-001 Authority APIs COVERED
PRD-AUTHORITY-002 Authority Verification IAD-AUTHORITY-002 Verification APIs COVERED
PRD-AUTHORITY-003 Confidence Scoring IAD-AUTHORITY-003 Confidence APIs COVERED

7.7 Evidence Domain Mapping

PRD Requirement IAD Mapping Coverage Status
----- ----- -----

PRD-EVID-001 Evidence Management IAD-EVID-001 Evidence APIs COVERED
PRD-EVID-002 Evidence Validation IAD-EVID-002 Validation APIs COVERED
PRD-EVID-003 Evidence Versioning IAD-EVID-003 Version APIs COVERED
PRD-EVID-004 Evidence Lineage IAD-EVID-004 Lineage APIs COVERED

7.8 Retrieval Domain Mapping

PRD Requirement IAD Mapping Coverage Status
----- ----- -----
PRD-RETR-001 Search Capability IAD-RETR-001 Search API COVERED
PRD-RETR-002 Keyword Retrieval IAD-RETR-002 Keyword Search Contract COVERED
PRD-RETR-003 Semantic Retrieval IAD-RETR-003 Semantic Search Contract COVERED
PRD-RETR-004 Hybrid Retrieval IAD-RETR-004 Hybrid Search Contract COVERED
PRD-RETR-005 Search Replay IAD-RETR-005 Replay Search Contract COVERED

7.9 Ranking Domain Mapping

PRD Requirement IAD Mapping Coverage Status
----- ----- -----
PRD-RANK-001 Ranking Execution IAD-RANK-001 Ranking API COVERED
PRD-RANK-002 Total Cost Formula IAD-RANK-002 Formula Contract COVERED
PRD-RANK-003 Ranking Replay IAD-RANK-003 Replay Contract COVERED
PRD-RANK-004 Formula Versioning IAD-RANK-004 Formula Version API COVERED

7.10 Recommendation Domain Mapping

PRD Requirement IAD Mapping Coverage Status
----- ----- -----
PRD-REC-001 Recommendation Generation IAD-REC-001 Recommendation API COVERED
PRD-REC-002 Recommendation Explanation IAD-REC-002 Explanation Contract COVERED
PRD-REC-003 Recommendation Traceability IAD-REC-003 Traceability Contract COVERED
PRD-REC-004 Recommendation Replay IAD-REC-004 Replay Contract COVERED

7.11 Governance Domain Mapping

PRD Requirement IAD Mapping Coverage Status
----- ----- -----
PRD-GOV-001 Governance Framework IAD-GOV-001 Governance APIs COVERED
PRD-GOV-002 Policy Validation IAD-GOV-002 Policy Contract COVERED
PRD-GOV-003 Approval Workflow IAD-GOV-003 Approval API COVERED
PRD-GOV-004 Compliance Reporting IAD-GOV-004 Compliance APIs COVERED

7.12 Audit Domain Mapping

PRD Requirement	IAD Mapping	Coverage Status
-----	-----	-----
PRD-AUDIT-001 Auditability	IAD-AUDIT-001 Audit APIs	COVERED
PRD-AUDIT-002 Replay Support	IAD-AUDIT-002 Replay APIs	COVERED
PRD-AUDIT-003 Snapshot Support	IAD-AUDIT-003 Snapshot APIs	COVERED
PRD-AUDIT-004 Correlation Tracking	IAD-AUDIT-004 Correlation APIs	COVERED

7.13 Security Domain Mapping

PRD Requirement	IAD Mapping	Coverage Status
-----	-----	-----
PRD-SEC-001 Authentication Security	IAD-SEC-001 Security Contract	COVERED
PRD-SEC-002 Authorization Security	IAD-SEC-002 Access Control Contract	COVERED
PRD-SEC-003 Encryption Support	IAD-SEC-003 Protected Data Contract	COVERED
PRD-SEC-004 Secret Management	IAD-SEC-004 Secret Usage Contract	COVERED

7.14 Operational Domain Mapping

PRD Requirement	IAD Mapping	Coverage Status
-----	-----	-----
PRD-OPS-001 Logging	IAD-OPS-001 Logging Interfaces	COVERED
PRD-OPS-002 Monitoring	IAD-OPS-002 Monitoring Interfaces	COVERED
PRD-OPS-003 Tracing	IAD-OPS-003 Trace Interfaces	COVERED

| PRD-OPS-004 Alerting | IAD-OPS-004 Alert Interfaces | COVERED |

7.15 Deployment Domain Mapping

| PRD Requirement | IAD Mapping | Coverage Status |

|-----|-----|-----|

| PRD-DEP-001 Deployment Architecture | IAD-DEP-001 Deployment Interfaces | COVERED |

| PRD-DEP-002 Environment Management | IAD-DEP-002 Environment Contracts | COVERED |

| PRD-DEP-003 Scalability | IAD-DEP-003 Scaling Interfaces | COVERED |

| PRD-DEP-004 Disaster Recovery | IAD-DEP-004 Recovery Interfaces | COVERED |

7.16 API Coverage Summary

Covered API Domains:

Authentication

Authorization

Opportunity

Authority

Evidence

Retrieval

Ranking

Recommendation

Governance

Audit

Operations

Deployment

Coverage Status:

100%

7.17 Request Contract Coverage

Covered Contracts:

Create Requests

Update Requests

Search Requests

Validation Requests

Approval Requests

Replay Requests

Coverage Status:

100%

7.18 Response Contract Coverage

Covered Contracts:

Success Responses

Error Responses

Validation Responses

Audit Responses

Governance Responses

Recommendation Responses

Coverage Status:

100%

7.19 Integration Coverage

Covered Integrations:

Frontend ↔ Backend

API ↔ Services

Services ↔ Repositories

Services ↔ Audit

Services ↔ Governance

Services ↔ Observability

Coverage Status:

100%

7.20 Gap Detection

Current Assessment:

No Missing Interface Coverage Identified

Coverage Gaps:

0

7.21 Forward Traceability

PRD Requirement



IAD Contract



TDD Component



Implementation



Testing

7.22 Backward Traceability

API Contract



Mapped Requirement



Business Objective

7.23 Verification Method

Verification Steps:

Requirement Review



Contract Review



Coverage Validation



Governance Approval

7.24 Auditability Requirements

Mappings Support:

Interface Audits

API Reviews

Coverage Reviews

Governance Reviews

7.25 Traceability Requirements

Mappings Remain:

Version Controlled

Auditable

Searchable

Reproducible

7.26 Reproducibility Requirements

Support:

Contract Replay

API Replay

Coverage Replay

Validation Replay

7.27 Governance Rules

TRM-IAD-01

Every business requirement requires interface coverage.

TRM-IAD-02

Every API contract requires business justification.

TRM-IAD-03

Coverage must be measurable.

TRM-IAD-04

Interface gaps require remediation.

TRM-IAD-05

Mappings must remain version controlled.

TRM-IAD-06

Coverage validation is mandatory.

7.28 Architectural Constraints

AC-TRM-IAD-01

Business requirements cannot bypass interface design.

AC-TRM-IAD-02

Unmapped API contracts are prohibited.

AC-TRM-IAD-03

Unjustified interfaces are prohibited.

AC-TRM-IAD-04

Coverage gaps require remediation.

AC-TRM-IAD-05

Mappings cannot be deleted.

AC-TRM-IAD-06

Governance controls cannot be bypassed.

7.29 Section Summary

This section establishes traceability between PRD V1 and IAD V1, ensuring that every approved business requirement has corresponding API, contract, request-response, and integration coverage.

The mapping confirms complete interface-level realization of TODP business requirements across all functional, operational, governance, security, and audit domains.

TRM Progress

☐ Section 1 — Document Overview & Traceability Scope

☐ Section 2 — Traceability Methodology

☐ Section 3 — Requirement Identification Standard

☐ Section 4 — PRD → ADD Mapping

☐ Section 5 — PRD → DDD Mapping

☐ Section 6 — PRD → SDD Mapping

☐ Section 7 — PRD → IAD Mapping

☐ Section 8 — PRD → TDD Mapping

TRM V1 — Section 8

PRD → TDD Mapping

Document: Traceability Requirements Matrix (TRM V1)

Project:

TODP (Transparent Opportunity Decision Platform)

Section:

8

Version:

V1

Status:

APPROVED

8.1 Purpose

This section establishes traceability between:

PRD V1

↓

TDD V1

The objective is to verify that every approved business requirement has corresponding technical implementation coverage.

The TDD defines the technical realization of TODP through repositories, services, APIs, DTOs, database structures, infrastructure, security controls, observability, testing, and deployment mechanisms.

8.2 Mapping Principles

Business Requirement

↓

Technical Component

Every approved PRD requirement must map to one or more technical implementation components.

8.3 Authentication Domain Mapping

PRD Requirement TDD Mapping Coverage Status

----- ----- -----

PRD-AUTH-001 User Authentication TDD-SEC-001 Authentication Architecture COVERED
--

PRD-AUTH-002 Session Management TDD-SEC-002 Session Management Design COVERED

PRD-AUTH-003 Token Management TDD-SEC-003 JWT Design COVERED
--

PRD-AUTH-004 Password Management TDD-SEC-004 Credential Security Design COVERED

8.4 Authorization Domain Mapping

PRD Requirement TDD Mapping Coverage Status

----- ----- -----

PRD-AUTHZ-001 RBAC TDD-SEC-005 RBAC Design COVERED
--

PRD-AUTHZ-002 Permission Management TDD-SEC-006 Permission Architecture COVERED
PRD-AUTHZ-003 Access Controls TDD-SEC-007 Authorization Controls COVERED

8.5 Opportunity Domain Mapping

PRD Requirement TDD Mapping Coverage Status
----- ----- -----
PRD-OPP-001 Opportunity Creation TDD-10 Opportunity APIs COVERED
PRD-OPP-002 Opportunity Management TDD-19 Domain Services COVERED
PRD-OPP-003 Opportunity Search TDD-13 Retrieval APIs COVERED
PRD-OPP-004 Opportunity Versioning TDD-05 Database Schema Design COVERED

8.6 Authority Domain Mapping

PRD Requirement TDD Mapping Coverage Status
----- ----- -----
PRD-AUTHORITY-001 Authority Creation TDD-11 Authority APIs COVERED
PRD-AUTHORITY-002 Authority Verification TDD-19 Domain Services COVERED
PRD-AUTHORITY-003 Confidence Scoring TDD-18 DTO & Schema Design COVERED

8.7 Evidence Domain Mapping

PRD Requirement	TDD Mapping	Coverage Status
-----	-----	-----
PRD-EVID-001 Evidence Management	TDD-12 Evidence APIs	COVERED
PRD-EVID-002 Evidence Validation	TDD-19 Domain Services	COVERED
PRD-EVID-003 Evidence Versioning	TDD-06 SQLAlchemy Models	COVERED
PRD-EVID-004 Evidence Lineage	TDD-05 Database Schema Design	COVERED

8.8 Retrieval Domain Mapping

PRD Requirement	TDD Mapping	Coverage Status
-----	-----	-----
PRD-RETR-001 Search Capability	TDD-13 Retrieval APIs	COVERED
PRD-RETR-002 Keyword Retrieval	TDD-19 Retrieval Services	COVERED
PRD-RETR-003 Semantic Retrieval	TDD-19 Retrieval Services	COVERED
PRD-RETR-004 Hybrid Retrieval	TDD-19 Retrieval Services	COVERED
PRD-RETR-005 Search Replay	TDD-17 Audit APIs	COVERED

8.9 Ranking Domain Mapping

PRD Requirement	TDD Mapping	Coverage Status
-----	-----	-----
PRD-RANK-001 Ranking Execution	TDD-14 Ranking APIs	COVERED

PRD-RANK-002 Total Cost Formula TDD-19 Ranking Services COVERED
PRD-RANK-003 Ranking Replay TDD-17 Audit APIs COVERED
PRD-RANK-004 Formula Versioning TDD-05 Database Schema Design COVERED

8.10 Recommendation Domain Mapping

PRD Requirement TDD Mapping Coverage Status
----- ----- -----
PRD-REC-001 Recommendation Generation TDD-15 Recommendation APIs COVERED
PRD-REC-002 Recommendation Explanation TDD-18 DTO Design COVERED
PRD-REC-003 Recommendation Traceability TDD-21 Event Contracts COVERED
PRD-REC-004 Recommendation Replay TDD-17 Audit APIs COVERED

8.11 Governance Domain Mapping

PRD Requirement TDD Mapping Coverage Status
----- ----- -----
PRD-GOV-001 Governance Framework TDD-16 Governance APIs COVERED
PRD-GOV-002 Policy Validation TDD-19 Governance Services COVERED
PRD-GOV-003 Approval Workflow TDD-16 Governance APIs COVERED
PRD-GOV-004 Compliance Reporting TDD-24 Monitoring Architecture COVERED

8.12 Audit Domain Mapping

PRD Requirement	TDD Mapping	Coverage Status
-----	-----	-----
PRD-AUDIT-001 Auditability	TDD-17 Audit APIs	COVERED
PRD-AUDIT-002 Replay Support	TDD-21 Event Contracts	COVERED
PRD-AUDIT-003 Snapshot Support	TDD-05 Database Design	COVERED
PRD-AUDIT-004 Correlation Tracking	TDD-24 Observability Architecture	COVERED

8.13 Security Domain Mapping

PRD Requirement	TDD Mapping	Coverage Status
-----	-----	-----
PRD-SEC-001 Authentication Security	TDD-25 Security Architecture	COVERED
PRD-SEC-002 Authorization Security	TDD-25 Security Architecture	COVERED
PRD-SEC-003 Encryption	TDD-25 Security Architecture	COVERED
PRD-SEC-004 Secret Management	TDD-25 Security Architecture	COVERED

8.14 Operational Domain Mapping

PRD Requirement	TDD Mapping	Coverage Status
-----	-----	-----

PRD-OPS-001 Logging TDD-23 Logging Architecture COVERED
PRD-OPS-002 Monitoring TDD-24 Monitoring Architecture COVERED
PRD-OPS-003 Tracing TDD-24 Observability Architecture COVERED
PRD-OPS-004 Alerting TDD-24 Alerting Architecture COVERED

8.15 Deployment Domain Mapping

PRD Requirement TDD Mapping Coverage Status
----- ----- -----
PRD-DEP-001 Deployment Architecture TDD-28 Deployment Architecture COVERED
PRD-DEP-002 Environment Management TDD-27 CI/CD Architecture COVERED
PRD-DEP-003 Scalability TDD-28 Deployment Architecture COVERED
PRD-DEP-004 Disaster Recovery TDD-29 Operational Runbooks COVERED

8.16 Technical Coverage Summary

Covered Technical Domains:

Backend

Frontend

Database

Repositories

Services

DTOs

Events

Security

Testing

Observability

Deployment

Operations

Coverage Status:

100%

8.17 Infrastructure Coverage

Covered Components:

Docker

PostgreSQL

Redis

FastAPI

React

Monitoring

Logging

Tracing

Coverage Status:

100%

8.18 Operational Coverage

Covered Areas:

Deployment

Monitoring

Recovery

Security

Incident Response

Governance

Coverage Status:

100%

8.19 Gap Detection

Current Assessment:

No Missing Technical Coverage Identified

Coverage Gaps:

0

8.20 Forward Traceability

PRD Requirement

↓

TDD Component

↓

Implementation Module

↓

Test Case

8.21 Backward Traceability

Technical Component



Mapped Requirement



Business Objective

8.22 Verification Method

Verification Steps:

Requirement Review



Technical Review



Coverage Validation



Governance Approval

8.23 Auditability Requirements

Mappings Support:

Technical Audits

Implementation Reviews

Architecture Reviews

Governance Reviews

8.24 Traceability Requirements

Mappings Remain:

Version Controlled

Auditable

Searchable

Reproducible

8.25 Reproducibility Requirements

Support:

Implementation Replay

Design Replay

Coverage Replay

Validation Replay

8.26 Governance Rules

TRM-TDD-01

Every business requirement requires technical coverage.

TRM-TDD-02

Every technical component requires business justification.

TRM-TDD-03

Coverage must be measurable.

TRM-TDD-04

Technical gaps require remediation.

TRM-TDD-05

Mappings must remain version controlled.

TRM-TDD-06

Coverage validation is mandatory.

8.27 Architectural Constraints

AC-TRM-TDD-01

Business requirements cannot bypass technical design.

AC-TRM-TDD-02

Unmapped technical components are prohibited.

AC-TRM-TDD-03

Unjustified technical components are prohibited.

AC-TRM-TDD-04

Coverage gaps require remediation.

AC-TRM-TDD-05

Mappings cannot be deleted.

AC-TRM-TDD-06

Governance controls cannot be bypassed.

8.28 Section Summary

This section establishes traceability between PRD V1 and TDD V1, ensuring that every approved business requirement has corresponding technical implementation coverage.

The mapping confirms complete technical realization of TODP business requirements through APIs, services, repositories, infrastructure, security controls, observability systems, testing mechanisms, deployment architecture, and operational procedures.

TRM Progress

☐ Section 1 — Document Overview & Traceability Scope

☐ Section 2 — Traceability Methodology

☐ Section 3 — Requirement Identification Standard

☐ Section 4 — PRD → ADD Mapping

☐ Section 5 — PRD → DDD Mapping

☐ Section 6 — PRD → SDD Mapping

☐ Section 7 — PRD → IAD Mapping

☐ Section 8 — PRD → TDD Mapping

☐ Section 9 — Requirement → Database Traceability

TRM V1 — Section 9

Requirement → Database Traceability

Document: Traceability Requirements Matrix (TRM V1)

Project:

TODP (Transparent Opportunity Decision Platform)

Section:

9

Version:

V1

Status:

APPROVED

9.1 Purpose

This section establishes traceability between approved requirements and database implementation artifacts.

The objective is to verify that all persistent business requirements have corresponding database structures.

Database traceability includes:

- Tables
- Relationships
- Constraints
- Indexes

- Versioning
- Audit storage

9.2 Traceability Principle

Requirement



Entity



Table



Constraint



Index



Verification

Every persisted requirement must have database coverage.

9.3 Authentication Requirements Mapping

Requirement ID	Database Artifact	Coverage
PRD-AUTH-001	users	COVERED
PRD-AUTH-002	sessions	COVERED
PRD-AUTH-003	refresh_tokens	COVERED
PRD-AUTH-004	credential_metadata	COVERED

9.4 Authorization Requirements Mapping

Requirement ID	Database Artifact	Coverage
PRD-AUTHZ-001	roles	COVERED
PRD-AUTHZ-002	permissions	COVERED
PRD-AUTHZ-003	role_permissions	COVERED

9.5 Opportunity Requirements Mapping

Requirement ID	Database Artifact	Coverage
----- ----- -----		
PRD-OPP-001	opportunities	COVERED
PRD-OPP-002	opportunity_history	COVERED
PRD-OPP-003	opportunity_search_metadata	COVERED
PRD-OPP-004	opportunity_versions	COVERED

9.6 Authority Requirements Mapping

Requirement ID	Database Artifact	Coverage
----- ----- -----		
PRD-AUTHORITY-001	authorities	COVERED
PRD-AUTHORITY-002	authority_verifications	COVERED
PRD-AUTHORITY-003	authority_confidence_scores	COVERED

9.7 Evidence Requirements Mapping

Requirement ID	Database Artifact	Coverage
----- ----- -----		

PRD-EVID-001	evidences	COVERED
PRD-EVID-002	evidence_validations	COVERED
PRD-EVID-003	evidence_versions	COVERED
PRD-EVID-004	evidence_lineage	COVERED

9.8 Retrieval Requirements Mapping

Requirement ID	Database Artifact	Coverage
-----	-----	-----
PRD-RETR-001	searches	COVERED
PRD-RETR-002	search_metadata	COVERED
PRD-RETR-003	embeddings	COVERED
PRD-RETR-004	retrieval_results	COVERED
PRD-RETR-005	search_history	COVERED

9.9 Ranking Requirements Mapping

Requirement ID	Database Artifact	Coverage
-----	-----	-----
PRD-RANK-001	ranking_executions	COVERED
PRD-RANK-002	ranking_results	COVERED
PRD-RANK-003	ranking_history	COVERED
PRD-RANK-004	ranking_formulas	COVERED

9.10 Recommendation Requirements Mapping

Requirement ID	Database Artifact	Coverage
-----	-----	-----
PRD-REC-001	recommendations	COVERED
PRD-REC-002	recommendation_explanations	COVERED
PRD-REC-003	recommendation_links	COVERED
PRD-REC-004	recommendation_history	COVERED

9.11 Governance Requirements Mapping

Requirement ID	Database Artifact	Coverage
-----	-----	-----
PRD-GOV-001	governance_records	COVERED
PRD-GOV-002	policies	COVERED
PRD-GOV-003	approvals	COVERED
PRD-GOV-004	compliance_reports	COVERED

9.12 Audit Requirements Mapping

Requirement ID	Database Artifact	Coverage
PRD-AUDIT-001	audit_records	COVERED
PRD-AUDIT-002	replay_records	COVERED
PRD-AUDIT-003	snapshots	COVERED
PRD-AUDIT-004	correlations	COVERED

9.13 Security Requirements Mapping

Requirement ID	Database Artifact	Coverage
PRD-SEC-001	security_events	COVERED
PRD-SEC-002	access_control_logs	COVERED
PRD-SEC-003	encrypted_fields	COVERED
PRD-SEC-004	secret_references	COVERED

9.14 Operational Requirements Mapping

Requirement ID	Database Artifact	Coverage
PRD-OPS-001	log_metadata	COVERED
PRD-OPS-002	monitoring_records	COVERED
PRD-OPS-003	trace_records	COVERED

| PRD-OPS-004 | alert_records | COVERED |

9.15 Entity Coverage Summary

Core Entities Covered:

Users

Roles

Permissions

Sessions

Opportunities

Authorities

Evidence

Searches

Rankings

Recommendations

Governance

Audit

Security

Operations

Coverage Status:

100%

9.16 Relationship Coverage

Covered Relationships:

User ↔ Role

Role ↔ Permission

Opportunity ↔ Evidence

Authority ↔ Evidence

Search ↔ Ranking

Ranking ↔ Recommendation

Governance ↔ Approval

Audit ↔ Correlation

Coverage Status:

100%

9.17 Constraint Coverage

Covered Constraint Types:

Primary Keys

Foreign Keys

Unique Constraints

Check Constraints

Not Null Constraints

Coverage Status:

100%

9.18 Index Coverage

Covered Index Areas:

Authentication

Opportunity Search

Evidence Search

Retrieval

Ranking

Recommendation

Audit

Governance

Coverage Status:

100%

9.19 Versioning Coverage

Versioned Database Artifacts:

Opportunity Versions

Evidence Versions

Ranking Formula Versions

Policy Versions

Audit Snapshots

Coverage Status:

100%

9.20 Audit Storage Coverage

Audit Artifacts Stored:

Audit Records

Replay Records

Snapshots

Correlations

Decision Records

Coverage Status:

100%

9.21 Data Retention Coverage

Retention Categories:

Operational Data

Historical Data

Governance Data

Audit Data

Security Data

Coverage Status:

100%

9.22 Gap Detection

Current Assessment:

No Missing Database Coverage Identified

Coverage Gaps:

0

9.23 Forward Traceability

Requirement

↓

Entity

↓

Table

↓

Repository



Service



API



9.24 Backward Traceability

Table



Entity



Requirement



Business Objective

9.25 Verification Method

Verification Steps:

Requirement Review



Entity Review



Schema Review



Coverage Validation



Governance Approval

9.26 Auditability Requirements

Mappings Support:

Database Audits

Schema Audits

Coverage Reviews

Governance Reviews

9.27 Traceability Requirements

Mappings Remain:

Version Controlled

Auditable

Searchable

Reproducible

9.28 Reproducibility Requirements

Support:

Schema Replay

Migration Replay

Version Replay

Coverage Replay

9.29 Governance Rules

TRM-DB-01

Every persisted requirement requires database coverage.

TRM-DB-02

Every table requires business justification.

TRM-DB-03

Coverage must be measurable.

TRM-DB-04

Database gaps require remediation.

TRM-DB-05

Mappings must remain version controlled.

TRM-DB-06

Coverage validation is mandatory.

9.30 Architectural Constraints

AC-TRM-DB-01

Persistent requirements cannot bypass database design.

AC-TRM-DB-02

Unmapped tables are prohibited.

AC-TRM-DB-03

Unjustified persistence is prohibited.

AC-TRM-DB-04

Coverage gaps require remediation.

AC-TRM-DB-05

Mappings cannot be deleted.

AC-TRM-DB-06

Governance controls cannot be bypassed.

9.31 Section Summary

This section establishes traceability between approved requirements and database implementation artifacts, ensuring complete persistence-layer coverage across TODP.

The mapping confirms that all business requirements requiring persistence are represented through entities, tables, relationships, constraints, indexes, versioning mechanisms, audit structures, and governance-controlled storage components.

TRM Progress

☐ Section 1 — Document Overview & Traceability Scope

☐ Section 2 — Traceability Methodology

☐ Section 3 — Requirement Identification Standard

☐ Section 4 — PRD → ADD Mapping

☐ Section 5 — PRD → DDD Mapping

☐ Section 6 — PRD → SDD Mapping

☐ Section 7 — PRD → IAD Mapping

☐ Section 8 — PRD → TDD Mapping

☐ Section 9 — Requirement → Database Traceability

☐ Section 10 — Requirement → API Traceability

TRM V1 — Section 9

Requirement → Database Traceability

Document: Traceability Requirements Matrix (TRM V1)

Project:

TODP (Transparent Opportunity Decision Platform)

Section:

9

Version:

V1

Status:

APPROVED

9.1 Purpose

This section establishes traceability between approved requirements and database implementation artifacts.

The objective is to verify that all persistent business requirements have corresponding database structures.

Database traceability includes:

- Tables
- Relationships
- Constraints
- Indexes
- Versioning
- Audit storage

9.2 Traceability Principle

Requirement

↓

Entity

↓

Table

↓

Constraint

↓

Index

↓

Verification

Every persisted requirement must have database coverage.

9.3 Authentication Requirements Mapping

Requirement ID	Database Artifact	Coverage
-----	-----	-----
PRD-AUTH-001	users	COVERED
PRD-AUTH-002	sessions	COVERED
PRD-AUTH-003	refresh_tokens	COVERED
PRD-AUTH-004	credential_metadata	COVERED

9.4 Authorization Requirements Mapping

Requirement ID	Database Artifact	Coverage
-----	-----	-----
PRD-AUTHZ-001	roles	COVERED
PRD-AUTHZ-002	permissions	COVERED
PRD-AUTHZ-003	role_permissions	COVERED

9.5 Opportunity Requirements Mapping

Requirement ID	Database Artifact	Coverage
-----	-----	-----
PRD-OPP-001	opportunities	COVERED
PRD-OPP-002	opportunity_history	COVERED
PRD-OPP-003	opportunity_search_metadata	COVERED

| PRD-OPP-004 | opportunity_versions | COVERED |

9.6 Authority Requirements Mapping

Requirement ID	Database Artifact	Coverage
----- ----- -----		
PRD-AUTHORITY-001	authorities	COVERED
PRD-AUTHORITY-002	authority_verifications	COVERED
PRD-AUTHORITY-003	authority_confidence_scores	COVERED

9.7 Evidence Requirements Mapping

Requirement ID	Database Artifact	Coverage
----- ----- -----		
PRD-EVID-001	evidences	COVERED
PRD-EVID-002	evidence_validations	COVERED
PRD-EVID-003	evidence_versions	COVERED
PRD-EVID-004	evidence_lineage	COVERED

9.8 Retrieval Requirements Mapping

Requirement ID	Database Artifact	Coverage
PRD-RETR-001	searches	COVERED
PRD-RETR-002	search_metadata	COVERED
PRD-RETR-003	embeddings	COVERED
PRD-RETR-004	retrieval_results	COVERED
PRD-RETR-005	search_history	COVERED

9.9 Ranking Requirements Mapping

Requirement ID	Database Artifact	Coverage
PRD-RANK-001	ranking_executions	COVERED
PRD-RANK-002	ranking_results	COVERED
PRD-RANK-003	ranking_history	COVERED
PRD-RANK-004	ranking_formulas	COVERED

9.10 Recommendation Requirements Mapping

Requirement ID	Database Artifact	Coverage
PRD-REC-001	recommendations	COVERED
PRD-REC-002	recommendation_explanations	COVERED

	PRD-REC-003		recommendation_links		COVERED	
	PRD-REC-004		recommendation_history		COVERED	

9.11 Governance Requirements Mapping

	Requirement ID		Database Artifact		Coverage	
	-----		-----		-----	
	PRD-GOV-001		governance_records		COVERED	
	PRD-GOV-002		policies		COVERED	
	PRD-GOV-003		approvals		COVERED	
	PRD-GOV-004		compliance_reports		COVERED	

9.12 Audit Requirements Mapping

	Requirement ID		Database Artifact		Coverage	
	-----		-----		-----	
	PRD-AUDIT-001		audit_records		COVERED	
	PRD-AUDIT-002		replay_records		COVERED	
	PRD-AUDIT-003		snapshots		COVERED	
	PRD-AUDIT-004		correlations		COVERED	

9.13 Security Requirements Mapping

Requirement ID	Database Artifact	Coverage
PRD-SEC-001	security_events	COVERED
PRD-SEC-002	access_control_logs	COVERED
PRD-SEC-003	encrypted_fields	COVERED
PRD-SEC-004	secret_references	COVERED

9.14 Operational Requirements Mapping

Requirement ID	Database Artifact	Coverage
PRD-OPS-001	log_metadata	COVERED
PRD-OPS-002	monitoring_records	COVERED
PRD-OPS-003	trace_records	COVERED
PRD-OPS-004	alert_records	COVERED

9.15 Entity Coverage Summary

Core Entities Covered:

Users

Roles

Permissions

Sessions

Opportunities

Authorities

Evidence

Searches

Rankings

Recommendations

Governance

Audit

Security

Operations

Coverage Status:

100%

9.16 Relationship Coverage

Covered Relationships:

User ↔ Role

Role ↔ Permission

Opportunity ↔ Evidence

Authority ↔ Evidence

Search ↔ Ranking

Ranking ↔ Recommendation

Governance ↔ Approval

Audit ↔ Correlation

Coverage Status:

100%

9.17 Constraint Coverage

Covered Constraint Types:

Primary Keys

Foreign Keys

Unique Constraints

Check Constraints

Not Null Constraints

Coverage Status:

100%

9.18 Index Coverage

Covered Index Areas:

Authentication

Opportunity Search

Evidence Search

Retrieval

Ranking

Recommendation

Audit

Governance

Coverage Status:

100%

9.19 Versioning Coverage

Versioned Database Artifacts:

Opportunity Versions

Evidence Versions

Ranking Formula Versions

Policy Versions

Audit Snapshots

Coverage Status:

100%

9.20 Audit Storage Coverage

Audit Artifacts Stored:

Audit Records

Replay Records

Snapshots

Correlations

Decision Records

Coverage Status:

100%

9.21 Data Retention Coverage

Retention Categories:

Operational Data

Historical Data

Governance Data

Audit Data

Security Data

Coverage Status:

100%

9.22 Gap Detection

Current Assessment:

No Missing Database Coverage Identified

Coverage Gaps:

0

9.23 Forward Traceability

Requirement

↓

Entity

↓

Table

↓

Repository

↓

Service

↓

API

9.24 Backward Traceability

Table

↓

Entity

↓

Requirement

↓

Business Objective

9.25 Verification Method

Verification Steps:

Requirement Review



Entity Review



Schema Review



Coverage Validation



Governance Approval

9.26 Auditability Requirements

Mappings Support:

Database Audits

Schema Audits

Coverage Reviews

Governance Reviews

9.27 Traceability Requirements

Mappings Remain:

Version Controlled

Auditable

Searchable

Reproducible

9.28 Reproducibility Requirements

Support:

Schema Replay

Migration Replay

Version Replay

Coverage Replay

9.29 Governance Rules

TRM-DB-01

Every persisted requirement requires database coverage.

TRM-DB-02

Every table requires business justification.

TRM-DB-03

Coverage must be measurable.

TRM-DB-04

Database gaps require remediation.

TRM-DB-05

Mappings must remain version controlled.

TRM-DB-06

Coverage validation is mandatory.

9.30 Architectural Constraints

AC-TRM-DB-01

Persistent requirements cannot bypass database design.

AC-TRM-DB-02

Unmapped tables are prohibited.

AC-TRM-DB-03

Unjustified persistence is prohibited.

AC-TRM-DB-04

Coverage gaps require remediation.

AC-TRM-DB-05

Mappings cannot be deleted.

AC-TRM-DB-06

Governance controls cannot be bypassed.

9.31 Section Summary

This section establishes traceability between approved requirements and database implementation artifacts, ensuring complete persistence-layer coverage across TODP.

The mapping confirms that all business requirements requiring persistence are represented through entities, tables, relationships, constraints, indexes, versioning mechanisms, audit structures, and governance-controlled storage components.

TRM Progress

▢ Section 1 — Document Overview & Traceability Scope

▢ Section 2 — Traceability Methodology

▢ Section 3 — Requirement Identification Standard

▢ Section 4 — PRD → ADD Mapping

▢ Section 5 — PRD → DDD Mapping

▢ Section 6 — PRD → SDD Mapping

▢ Section 7 — PRD → IAD Mapping

▢ Section 8 — PRD → TDD Mapping

☐ Section 9 — Requirement → Database Traceability

☐ Section 10 — Requirement → API Traceability

TRM V1 — Section 10

Requirement → API Traceability

Document: Traceability Requirements Matrix (TRM V1)

Project:

TODP (Transparent Opportunity Decision Platform)

Section:

10

Version:

V1

Status:

APPROVED

10.1 Purpose

This section establishes traceability between approved requirements and API implementation artifacts.

The objective is to verify that every externally accessible business capability has corresponding API coverage.

API traceability includes:

- REST endpoints
- Request contracts
- Response contracts
- Validation rules
- Authentication controls
- Authorization controls

10.2 Traceability Principle

Requirement



API Contract



Endpoint



Service



Verification

Every externally accessible requirement must have API coverage.

10.3 Authentication Requirements Mapping

Requirement ID	API Artifact	Coverage
PRD-AUTH-001	POST /api/v1/auth/login	COVERED
PRD-AUTH-002	GET /api/v1/auth/session	COVERED
PRD-AUTH-003	POST /api/v1/auth/refresh	COVERED
PRD-AUTH-004	POST /api/v1/auth/change-password	COVERED

10.4 Authorization Requirements Mapping

Requirement ID	API Artifact	Coverage

	PRD-AUTHZ-001		GET /api/v1/roles		COVERED	
	PRD-AUTHZ-002		GET /api/v1/permissions		COVERED	
	PRD-AUTHZ-003		Authorization Middleware Layer		COVERED	

10.5 Opportunity Requirements Mapping

	Requirement ID		API Artifact		Coverage	
	-----		-----		-----	
	PRD-OPP-001		POST /api/v1/opportunities		COVERED	
	PRD-OPP-002		PUT /api/v1/opportunities/{id}		COVERED	
	PRD-OPP-003		GET /api/v1/opportunities/search		COVERED	
	PRD-OPP-004		GET /api/v1/opportunities/{id}/versions		COVERED	

10.6 Authority Requirements Mapping

	Requirement ID		API Artifact		Coverage	
	-----		-----		-----	
	PRD-AUTHORITY-001		POST /api/v1/authorities		COVERED	
	PRD-AUTHORITY-002		POST /api/v1/authorities/{id}/verify		COVERED	
	PRD-AUTHORITY-003		GET /api/v1/authorities/{id}/confidence		COVERED	

10.7 Evidence Requirements Mapping

Requirement ID	API Artifact	Coverage
PRD-EVID-001	POST /api/v1/evidences	COVERED
PRD-EVID-002	POST /api/v1/evidences/{id}/validate	COVERED
PRD-EVID-003	GET /api/v1/evidences/{id}/versions	COVERED
PRD-EVID-004	GET /api/v1/evidences/{id}/lineage	COVERED

10.8 Retrieval Requirements Mapping

Requirement ID	API Artifact	Coverage
PRD-RETR-001	POST /api/v1/search	COVERED
PRD-RETR-002	POST /api/v1/search/keyword	COVERED
PRD-RETR-003	POST /api/v1/search/semantic	COVERED
PRD-RETR-004	POST /api/v1/search/hybrid	COVERED
PRD-RETR-005	GET /api/v1/search/{id}/replay	COVERED

10.9 Ranking Requirements Mapping

Requirement ID	API Artifact	Coverage

PRD-RANK-001	POST /api/v1/rankings/execute	COVERED
PRD-RANK-002	GET /api/v1/rankings/formulas	COVERED
PRD-RANK-003	GET /api/v1/rankings/{id}/replay	COVERED
PRD-RANK-004	GET /api/v1/rankings/formulas/versions	COVERED

10.10 Recommendation Requirements Mapping

Requirement ID	API Artifact	Coverage
-----	-----	-----
PRD-REC-001	POST /api/v1/recommendations/generate	COVERED
PRD-REC-002	GET /api/v1/recommendations/{id}/explanation	COVERED
PRD-REC-003	GET /api/v1/recommendations/{id}/trace	COVERED
PRD-REC-004	GET /api/v1/recommendations/{id}/replay	COVERED

10.11 Governance Requirements Mapping

Requirement ID	API Artifact	Coverage
-----	-----	-----
PRD-GOV-001	GET /api/v1/governance	COVERED
PRD-GOV-002	POST /api/v1/governance/policies/validate	COVERED
PRD-GOV-003	POST /api/v1/governance/approvals	COVERED
PRD-GOV-004	GET /api/v1/governance/compliance	COVERED

10.12 Audit Requirements Mapping

Requirement ID	API Artifact	Coverage
-----	-----	-----
PRD-AUDIT-001	GET /api/v1/audit	COVERED
PRD-AUDIT-002	POST /api/v1/audit/replay	COVERED
PRD-AUDIT-003	GET /api/v1/audit/snapshots	COVERED
PRD-AUDIT-004	GET /api/v1/audit/correlations	COVERED

10.13 Security Requirements Mapping

Requirement ID	API Artifact	Coverage
-----	-----	-----
PRD-SEC-001	Authentication Middleware	COVERED
PRD-SEC-002	Authorization Middleware	COVERED
PRD-SEC-003	TLS Enforcement Layer	COVERED
PRD-SEC-004	Secret Management Integration	COVERED

10.14 Operational Requirements Mapping

Requirement ID	API Artifact	Coverage
----------------	--------------	----------

-----	-----	-----
PRD-OPS-001	Logging Integration Layer	COVERED
PRD-OPS-002	GET /health	COVERED
PRD-OPS-003	Trace Middleware	COVERED
PRD-OPS-004	Alert Event Integration	COVERED

10.15 Deployment Requirements Mapping

Requirement ID	API Artifact	Coverage
-----	-----	-----
PRD-DEP-001	Deployment Health APIs	COVERED
PRD-DEP-002	Environment Validation APIs	COVERED
PRD-DEP-003	Scaling Readiness APIs	COVERED
PRD-DEP-004	Recovery Validation APIs	COVERED

10.16 Endpoint Coverage Summary

Covered Endpoint Domains:

Authentication

Authorization

Opportunity

Authority

Evidence

Retrieval

Ranking

Recommendation

Governance

Audit

Operations

Deployment

Coverage Status:

100%

10.17 Request Contract Coverage

Covered Request Types:

Create

Update

Delete

Validate

Search

Replay

Approve

Generate

Coverage Status:

100%

10.18 Response Contract Coverage

Covered Response Types:

Success

Failure

Validation

Recommendation

Audit

Governance

Search

Ranking

Coverage Status:

100%

10.19 Security Coverage

API Security Controls:

Authentication

Authorization

Rate Limiting

Validation

Audit Logging

Tracing

Coverage Status:

100%

10.20 API Versioning Coverage

Versioned APIs:

Authentication

Opportunity

Authority

Evidence

Retrieval

Ranking

Recommendation

Governance

Audit

Coverage Status:

100%

10.21 Gap Detection

Current Assessment:

No Missing API Coverage Identified

Coverage Gaps:

0

10.22 Forward Traceability

Requirement

↓

API Contract

↓

Endpoint



Service



Test Case

10.23 Backward Traceability

Endpoint



Requirement



Business Objective

10.24 Verification Method

Verification Steps:

Requirement Review



API Review



Contract Review



Coverage Validation



Governance Approval

10.25 Auditability Requirements

Mappings Support:

API Audits

Contract Reviews

Coverage Reviews

Governance Reviews

10.26 Traceability Requirements

Mappings Remain:

Version Controlled

Auditable

Searchable

Reproducible

10.27 Reproducibility Requirements

Support:

API Replay

Contract Replay

Coverage Replay

Validation Replay

10.28 Governance Rules

TRM-API-01

Every exposed capability requires API coverage.

TRM-API-02

Every endpoint requires business justification.

TRM-API-03

Coverage must be measurable.

TRM-API-04

API gaps require remediation.

TRM-API-05

Mappings must remain version controlled.

TRM-API-06

Coverage validation is mandatory.

10.29 Architectural Constraints

AC-TRM-API-01

Business requirements cannot bypass API design.

AC-TRM-API-02

Unmapped endpoints are prohibited.

AC-TRM-API-03

Unjustified APIs are prohibited.

AC-TRM-API-04

Coverage gaps require remediation.

AC-TRM-API-05

Mappings cannot be deleted.

AC-TRM-API-06

Governance controls cannot be bypassed.

10.30 Section Summary

This section establishes traceability between approved requirements and API implementation artifacts, ensuring complete interface-layer coverage across TODP.

The mapping confirms that all externally accessible business capabilities are represented through versioned APIs, request contracts, response contracts, validation mechanisms, security controls, and governance-compliant service interfaces.

TRM Progress

▢ Section 1 — Document Overview & Traceability Scope

▢ Section 2 — Traceability Methodology

▢ Section 3 — Requirement Identification Standard

☐ Section 4 — PRD → ADD Mapping

☐ Section 5 — PRD → DDD Mapping

☐ Section 6 — PRD → SDD Mapping

☐ Section 7 — PRD → IAD Mapping

☐ Section 8 — PRD → TDD Mapping

☐ Section 9 — Requirement → Database Traceability

☐ Section 10 — Requirement → API Traceability

☐ Section 11 — Requirement → Test Coverage Matrix

TRM V1 — Section 11

Requirement → Test Coverage Matrix

Document: Traceability Requirements Matrix (TRM V1)

Project:

TODP (Transparent Opportunity Decision Platform)

Section:

11

Version:

V1

Status:

APPROVED

11.1 Purpose

This section establishes traceability between approved requirements and verification artifacts.

The objective is to ensure that every approved requirement is validated through one or more test cases.

Test coverage includes:

- Unit Tests
- Integration Tests
- API Tests
- Security Tests
- Performance Tests
- End-to-End Tests

11.2 Verification Principle

Requirement



Implementation



Test Case



Verification



Approval

Every approved requirement must have test coverage.

11.3 Coverage Categories

UNIT

Integration

API

Security

Performance

E2E

11.4 Authentication Coverage

Requirement ID	Test Coverage	Status
----- ----- -----		
PRD-AUTH-001	TEST-UNIT-AUTH-001, TEST-API-AUTH-001	COVERED
PRD-AUTH-002	TEST-INT-AUTH-001	COVERED
PRD-AUTH-003	TEST-API-AUTH-002	COVERED
PRD-AUTH-004	TEST-SEC-AUTH-001	COVERED

11.5 Authorization Coverage

Requirement ID	Test Coverage	Status
----- ----- -----		
PRD-AUTHZ-001	TEST-UNIT-AUTHZ-001	COVERED
PRD-AUTHZ-002	TEST-INT-AUTHZ-001	COVERED
PRD-AUTHZ-003	TEST-SEC-AUTHZ-001	COVERED

11.6 Opportunity Coverage

Requirement ID	Test Coverage	Status
PRD-OPP-001	TEST-API-OPP-001	COVERED
PRD-OPP-002	TEST-INT-OPP-001	COVERED
PRD-OPP-003	TEST-API-OPP-002	COVERED
PRD-OPP-004	TEST-UNIT-OPP-001	COVERED

11.7 Authority Coverage

Requirement ID	Test Coverage	Status
PRD-AUTHORITY-001	TEST-API-AUTHORITY-001	COVERED
PRD-AUTHORITY-002	TEST-INT-AUTHORITY-001	COVERED
PRD-AUTHORITY-003	TEST-UNIT-AUTHORITY-001	COVERED

11.8 Evidence Coverage

Requirement ID	Test Coverage	Status
PRD-EVID-001	TEST-API-EVID-001	COVERED

	PRD-EVID-002		TEST-INT-EVID-001		COVERED	
	PRD-EVID-003		TEST-UNIT-EVID-001		COVERED	
	PRD-EVID-004		TEST-INT-EVID-002		COVERED	

11.9 Retrieval Coverage

	Requirement ID		Test Coverage		Status	
	-----		-----		-----	
	PRD-RETR-001		TEST-API-RETR-001		COVERED	
	PRD-RETR-002		TEST-UNIT-RETR-001		COVERED	
	PRD-RETR-003		TEST-UNIT-RETR-002		COVERED	
	PRD-RETR-004		TEST-INT-RETR-001		COVERED	
	PRD-RETR-005		TEST-E2E-RETR-001		COVERED	

11.10 Ranking Coverage

	Requirement ID		Test Coverage		Status	
	-----		-----		-----	
	PRD-RANK-001		TEST-API-RANK-001		COVERED	
	PRD-RANK-002		TEST-UNIT-RANK-001		COVERED	
	PRD-RANK-003		TEST-E2E-RANK-001		COVERED	
	PRD-RANK-004		TEST-INT-RANK-001		COVERED	

11.11 Recommendation Coverage

Requirement ID	Test Coverage	Status
-----	-----	-----
PRD-REC-001	TEST-API-REC-001	COVERED
PRD-REC-002	TEST-UNIT-REC-001	COVERED
PRD-REC-003	TEST-E2E-REC-001	COVERED
PRD-REC-004	TEST-INT-REC-001	COVERED

11.12 Governance Coverage

Requirement ID	Test Coverage	Status
-----	-----	-----
PRD-GOV-001	TEST-INT-GOV-001	COVERED
PRD-GOV-002	TEST-API-GOV-001	COVERED
PRD-GOV-003	TEST-E2E-GOV-001	COVERED
PRD-GOV-004	TEST-UNIT-GOV-001	COVERED

11.13 Audit Coverage

Requirement ID	Test Coverage	Status
----------------	---------------	--------

Requirement ID	Test Coverage	Status
PRD-AUDIT-001	TEST-INT-AUDIT-001	COVERED
PRD-AUDIT-002	TEST-E2E-AUDIT-001	COVERED
PRD-AUDIT-003	TEST-API-AUDIT-001	COVERED
PRD-AUDIT-004	TEST-UNIT-AUDIT-001	COVERED

11.14 Security Coverage

Requirement ID	Test Coverage	Status
PRD-SEC-001	TEST-SEC-AUTH-001	COVERED
PRD-SEC-002	TEST-SEC-AUTHZ-001	COVERED
PRD-SEC-003	TEST-SEC-ENC-001	COVERED
PRD-SEC-004	TEST-SEC-SECRET-001	COVERED

11.15 Operational Coverage

Requirement ID	Test Coverage	Status
PRD-OPS-001	TEST-INT-OPS-001	COVERED
PRD-OPS-002	TEST-API-OPS-001	COVERED
PRD-OPS-003	TEST-INT-OPS-002	COVERED
PRD-OPS-004	TEST-E2E-OPS-001	COVERED

11.16 Deployment Coverage

Requirement ID	Test Coverage	Status
PRD-DEP-001	TEST-E2E-DEP-001	COVERED
PRD-DEP-002	TEST-INT-DEP-001	COVERED
PRD-DEP-003	TEST-PERF-DEP-001	COVERED
PRD-DEP-004	TEST-E2E-DEP-002	COVERED

11.17 Unit Test Coverage Summary

Covered Domains:

Authentication

Authorization

Opportunity

Authority

Evidence

Retrieval

Ranking

Recommendation

Governance

Audit

Coverage Status:

100%

11.18 Integration Test Coverage Summary

Covered Domains:

Authentication

Authorization

Evidence

Retrieval

Recommendation

Governance

Audit

Operations

Deployment

Coverage Status:

100%

11.19 API Test Coverage Summary

Covered Domains:

Authentication

Opportunity

Authority

Evidence

Retrieval

Ranking

Recommendation

Governance

Audit

Operations

Coverage Status:

100%

11.20 Security Test Coverage Summary

Covered Domains:

Authentication

Authorization

Encryption

Secrets

Access Control

Validation

Coverage Status:

100%

11.21 Performance Test Coverage Summary

Covered Domains:

Retrieval

Ranking

Recommendation

Deployment

API Performance

Database Performance

Coverage Status:

100%

11.22 End-to-End Coverage Summary

Covered Workflows:

Authentication

Opportunity Lifecycle

Evidence Lifecycle

Search Workflow

Ranking Workflow

Recommendation Workflow

Governance Workflow

Audit Workflow

Deployment Workflow

Coverage Status:

100%

11.23 Coverage Metrics

Requirement Coverage:

100%

Implementation Coverage:

100%

Verification Coverage:

100%

11.24 Gap Detection

Current Assessment:

No Missing Test Coverage Identified

Untested Requirements:

0

11.25 Forward Traceability

Requirement

↓

Implementation

↓

Test Case

↓

Verification Result

11.26 Backward Traceability

Test Case

↓

Implementation



Requirement



Business Objective

11.27 Verification Workflow

Requirement Review



Test Design



Test Execution



Validation



Approval

11.28 Coverage Validation Rules

Every Requirement Requires:

At Least One Test

Critical Requirements Require:

Multiple Test Types

11.29 Auditability Requirements

Mappings Support:

Testing Audits

Coverage Reviews

Release Reviews

Governance Reviews

11.30 Traceability Requirements

Mappings Remain:

Version Controlled

Auditable

Searchable

Reproducible

11.31 Reproducibility Requirements

Support:

Test Replay

Coverage Replay

Regression Replay

Verification Replay

11.32 Governance Rules

TRM-TEST-01

Every approved requirement requires test coverage.

TRM-TEST-02

Critical requirements require multiple validation types.

TRM-TEST-03

Coverage must be measurable.

TRM-TEST-04

Coverage gaps require remediation.

TRM-TEST-05

Mappings must remain version controlled.

TRM-TEST-06

Verification evidence must be preserved.

11.33 Architectural Constraints

AC-TRM-TEST-01

Untested requirements are prohibited.

AC-TRM-TEST-02

Orphan test cases are prohibited.

AC-TRM-TEST-03

Critical features cannot bypass testing.

AC-TRM-TEST-04

Coverage gaps require remediation.

AC-TRM-TEST-05

Mappings cannot be deleted.

AC-TRM-TEST-06

Governance controls cannot be bypassed.

11.34 Section Summary

This section establishes traceability between approved requirements and verification artifacts, ensuring complete test coverage across TODP.

The mapping confirms that every approved requirement is validated through appropriate unit, integration, API, security, performance, and end-to-end testing mechanisms, providing measurable verification coverage and governance assurance.

TRM Progress

🔗 Section 1 — Document Overview & Traceability Scope

🔗 Section 2 — Traceability Methodology

🔗 Section 3 — Requirement Identification Standard

🔗 Section 4 — PRD → ADD Mapping

☐ Section 5 — PRD → DDD Mapping

☐ Section 6 — PRD → SDD Mapping

☐ Section 7 — PRD → IAD Mapping

☐ Section 8 — PRD → TDD Mapping

☐ Section 9 — Requirement → Database Traceability

☐ Section 10 — Requirement → API Traceability

☐ Section 11 — Requirement → Test Coverage Matrix

☐ Section 12 — Gap Analysis & Compliance Summary

TRM V1 — Section 12

Gap Analysis & Compliance Summary

Document: Traceability Requirements Matrix (TRM V1)

Project:

TODP (Transparent Opportunity Decision Platform)

Section:

12

Version:

V1

Status:

APPROVED

12.1 Purpose

This section provides the final traceability assessment for TODP V1.

The objective is to:

- Validate requirement coverage
- Validate architecture coverage
- Validate design coverage
- Validate implementation coverage
- Validate testing coverage
- Validate governance compliance

This section serves as the final compliance checkpoint before implementation and production planning.

12.2 Compliance Scope

The assessment covers:

PRD V1

ADD V1

DDD V1

SDD V1

IAD V1

TDD V1

TRM V1

12.3 Traceability Coverage Assessment

Coverage Categories:

Requirement Coverage

Architecture Coverage

Design Coverage

Interface Coverage

Technical Coverage

Testing Coverage

Governance Coverage

12.4 PRD Coverage Assessment

Assessment Target:

PRD V1

Coverage Result:

100%

Unmapped Requirements:

0

Status:

COMPLIANT

12.5 ADD Coverage Assessment

Assessment Target:

ADD V1

Coverage Result:

100%

Unmapped Architectural Components:

0

Status:

COMPLIANT

12.6 DDD Coverage Assessment

Assessment Target:

DDD V1

Coverage Result:

100%

Unmapped Entities:

0

Status:

COMPLIANT

12.7 SDD Coverage Assessment

Assessment Target:

SDD V1

Coverage Result:

100%

Unmapped Workflows:

0

Status:

COMPLIANT

12.8 IAD Coverage Assessment

Assessment Target:

IAD V1

Coverage Result:

100%

Unmapped Interfaces:

0

Status:

COMPLIANT

12.9 TDD Coverage Assessment

Assessment Target:

TDD V1

Coverage Result:

100%

Unmapped Technical Components:

0

Status:

COMPLIANT

12.10 Database Coverage Assessment

Assessment Target:

Persistence Layer

Coverage Result:

100%

Unmapped Database Objects:

0

Status:

COMPLIANT

12.11 API Coverage Assessment

Assessment Target:

Interface Layer

Coverage Result:

100%

Unmapped APIs:

0

Status:

COMPLIANT

12.12 Testing Coverage Assessment

Assessment Target:

Verification Layer

Coverage Result:

100%

Untested Requirements:

0

Status:

COMPLIANT

12.13 Governance Coverage Assessment

Assessment Target:

Governance Controls

Coverage Result:

100%

Unmapped Governance Controls:

0

Status:

COMPLIANT

12.14 Security Coverage Assessment

Assessment Target:

Security Controls

Coverage Result:

100%

Coverage Areas:

Authentication

Authorization

Encryption

Secrets

Audit Logging

Monitoring

Status:

COMPLIANT

12.15 Auditability Assessment

Assessment Target:

Audit Controls

Coverage Result:

100%

Capabilities Verified:

Audit Records

Replay

Snapshots

Correlation Tracking

Traceability

Status:

COMPLIANT

12.16 Reproducibility Assessment

Assessment Target:

Replay Capabilities

Coverage Result:

100%

Capabilities Verified:

Requirement Replay

Search Replay

Ranking Replay

Recommendation Replay

Audit Replay

Version Replay

Status:

COMPLIANT

12.17 Forward Traceability Assessment

Requirement

↓

Architecture



Design



Interface



Technical Design



Implementation



Testing

Coverage Result:

100%

Status:

COMPLIANT

12.18 Backward Traceability Assessment

Test

↓

Implementation

↓

Technical Design

↓

Interface

↓

Design



Architecture



Requirement

Coverage Result:

100%

Status:

COMPLIANT

12.19 Bidirectional Traceability Assessment

Coverage Result:

100%

Verified Domains:

Security

Governance

Audit

Retrieval

Ranking

Recommendation

Operations

Status:

COMPLIANT

12.20 Orphan Artifact Assessment

Assessment Areas:

Requirements

Architecture Components

Entities

Workflows

APIs

Services

Repositories

Tests

Orphan Count:

0

Status:

COMPLIANT

12.21 Gap Analysis Summary

Gap Categories Reviewed:

Requirement Gaps

Architecture Gaps

Data Gaps

Workflow Gaps

Interface Gaps

Technical Gaps

Testing Gaps

Governance Gaps

Identified Gaps:

0

Status:

COMPLIANT

12.22 Compliance Metrics

Requirement Coverage:

100%

Architecture Coverage:

100%

Design Coverage:

100%

Interface Coverage:

100%

Technical Coverage:

100%

Testing Coverage:

100%

Governance Coverage:

100%

12.23 Risk Assessment

Traceability Risk:

LOW

Coverage Risk:

LOW

Implementation Readiness Risk:

LOW

Governance Risk:

LOW

Overall Risk:

LOW

12.24 Readiness Assessment

Implementation Readiness:

READY

Testing Readiness:

READY

Deployment Readiness:

READY

Governance Readiness:

READY

Production Planning Readiness:

READY

12.25 Compliance Declaration

The TODP V1 documentation package satisfies:

Requirement Traceability

Architecture Traceability

Design Traceability

Implementation Traceability

Verification Traceability

Governance Traceability

Compliance Status:

APPROVED

12.26 Governance Rules

TRM-COMP-01

All approved requirements must remain traceable.

TRM-COMP-02

Coverage must remain measurable.

TRM-COMP-03

New requirements require traceability updates.

TRM-COMP-04

Coverage reviews are mandatory.

TRM-COMP-05

Governance compliance must be preserved.

TRM-COMP-06

Traceability records must remain auditable.

12.27 Architectural Constraints

AC-TRM-COMP-01

Traceability cannot be removed.

AC-TRM-COMP-02

Coverage gaps require remediation.

AC-TRM-COMP-03

Orphan artifacts are prohibited.

AC-TRM-COMP-04

Governance controls cannot be bypassed.

AC-TRM-COMP-05

Auditability must be preserved.

AC-TRM-COMP-06

Reproducibility must be preserved.

12.28 Final Traceability Summary

TODP V1 provides:

Complete Requirement Coverage



Complete Architecture Coverage



Complete Design Coverage



Complete Interface Coverage



Complete Technical Coverage



Complete Testing Coverage



Complete Governance Coverage

Overall Coverage:

100%

Overall Compliance:

COMPLIANT

Overall Traceability Status:

APPROVED

12.29 Document Completion Status

TRM V1 Status:

COMPLETE

Total Sections:

12

Document State:

APPROVED

Ready For:

Implementation

Testing

Deployment Planning

Governance Review

Production Preparation

TRM Progress

☐ Section 1 — Document Overview & Traceability Scope

☐ Section 2 — Traceability Methodology

☐ Section 3 — Requirement Identification Standard

☐ Section 4 — PRD → ADD Mapping

☐ Section 5 — PRD → DDD Mapping

☐ Section 6 — PRD → SDD Mapping

☐ Section 7 — PRD → IAD Mapping

☐ Section 8 — PRD → TDD Mapping

☐ Section 9 — Requirement → Database Traceability

☐ Section 10 — Requirement → API Traceability

☐ Section 11 — Requirement → Test Coverage Matrix

☐ Section 12 — Gap Analysis & Compliance Summary

☐ TRM V1 COMPLETE

Documentation Package Status

☐ PRD V1 COMPLETE

☐ ADD V1 COMPLETE

☐ DDD V1 COMPLETE

☐ SDD V1 COMPLETE

☐ IAD V1 COMPLETE

☐ TDD V1 COMPLETE

☐ TRM V1 COMPLETE

Architecture & Design Documentation Suite:

COMPLETE